JAIIB CAIIB

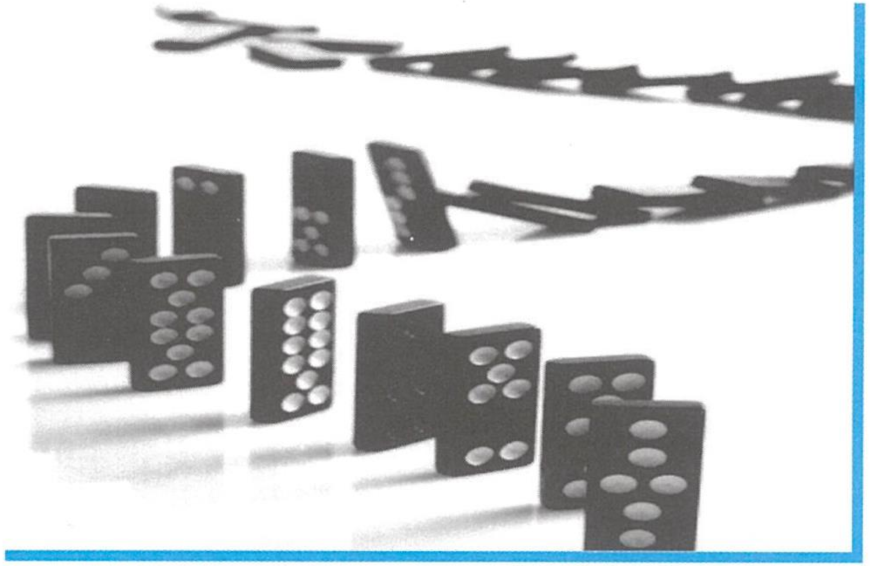
PREP

JAIIB CAIIB

PREP

@JAIIBCAIIBPREPOFFICIAL

MODULE D



BALANCE SHEET MANAGEMENT

UNIT 26 Components of Assets and Liabilities in Bank's Balance Sheet and Their Management

STRUCTURE

1. Preparation of Balance Sheet

Third Schedule: Form 'A'

Form of Balance Sheet

Balance Sheet as on 31st March, ...

Sec. 29 BR Act 1949

Capital and Liabilities	Schedule No.	₹
Capital	1	.....
Reserves and Surplus	2	.....
Minorities Interest	2A	
Deposits	3	.....
Borrowings	4	.....
Other Liabilities and Provisions	5	.....
Total		

SOF

Assets	Schedule No.	₹
Cash and Balance with RBI	6	.....
Balances with Banks and Money at Call and Short Notice	7	.....
Investments	8	.....
Advances	9	.....
Fixed Assets	10	.....
Other Assets	11	.....
Goodwill on consolidation[1]		.....
Debit balance of Profit and Loss account		
Total		.....
Contingent Liabilities	12	.....
Bills for collection		

AOF

Where there is more than one subsidiary and the aggregation results in Goodwill in some cases and C
Reserves in other cases, net effect to be shown in Schedule 2 and Assets side after giving separates r

17.8 PREPARATION OF PROFIT AND LOSS ACCOUNT

Form 'B' Third Schedule

Form of Profit and Loss Account

Profit and Loss Account for the Year Ended 31st March, 20......

	Schedule Number	Year Ended (₹)
I. Income:		
Interest Earned	13	....
Other Income	14	....
II. Expenditure:		
Interest Expended	15	....
Operating Expenses	16	....
Provisions and Contingencies		....
Total		....
III. Profit/Loss:		
Net Profit/(Loss) of the Year		....
Profit/(Loss) brought forward		....
Share of earnings/loss in Associates		....
Consolidated Net profit/(loss) for the year before deducting Minorities' Interest	17	....
Less: Minorities' Interest		
Consolidated profit/(loss) for the year attributable to the group		....
Add: Brought forward consolidated profit/(loss) attributable to the group		....
Total		....
IV. Appropriations:		
Transfer to Statutory Reserves		....
Transfer to other Reserves		
Transfer to Government/Proposed Dividend		....
Balance Carried over to Balance Sheet		....
Total		....
Earnings per Share[1]		....
Basic earnings per equity share		....
Diluted earnings per equity share		....

Earnings per share should be for both basic and diluted.

The summarised form and its components are:

Sources of Funds	Application of Funds
Capital	Cash In Hand and Balance with RBI
Reserves	Balances with Banks and Money at Call and Short Notice
Deposits	Investments
Borrowings	Advance
Other Liabilities and provisions	Fixed Assets
	Other Assets
TOTAL	TOTAL

Liability

Assets

FORM A

The Framework: Sources vs. Application of Funds

B

In terms of Section 29 of the Banking Regulation Act 1949.

L

Sources of Funds (Liabilities)

- Capital
- Reserves
- Deposits
- Borrowings
- Other Liabilities & Provisions

Application of Funds (Assets)

- Cash & Balances with RBI
- Balances with Banks/Money at Call
- Investments
- Advances
- Fixed Assets
- Other Assets

Liabilities: The Owners' Stake

Capital

Represents the owners' stake.

Serves as a cushion for depositors and creditors in case of losses.

Minimum requirement prescribed by RBI via Capital Adequacy Ratio (CAR).

Reserves and Surplus

Includes Statutory reserves, Capital reserves, Share premium, Revenue reserves, and the Balance in Profit and Loss account.

Liabilities: Public Funds & Borrowings

Deposits (Main Source)

- **Demand Deposits**: Current, overdue, call deposits.
- **Savings Deposits**: Savings bank accounts.
- **Term Deposits**: Fixed, short, recurring (repayable after period).

Borrowings

- Refinance from RBI & Commercial Banks.
- **Agencies**: IDBI, EXIM Bank of India, NABARD.

Other Liabilities & Contingent Obligations

Other Liabilities

Helvetica Now Display Bold, Deep Charcoal

- Bills Payable (Drafts, TTs)
- Inter-Office Adjustments (Net credit)
- Interest Accrued (Not yet due)

Contingent Liabilities (Off-Balance Sheet)

Obligations on behalf of constituents.

- Letters of Credit (LC)
- Guarantees & Acceptances
- Claims not acknowledged as debts
- Forward exchange contracts

Assets: Liquidity & Investments

Cash & Balances with RBI

Cash in hand (foreign/domestic) & Balances held for Statutory Cash Reserve Requirements (CRR).

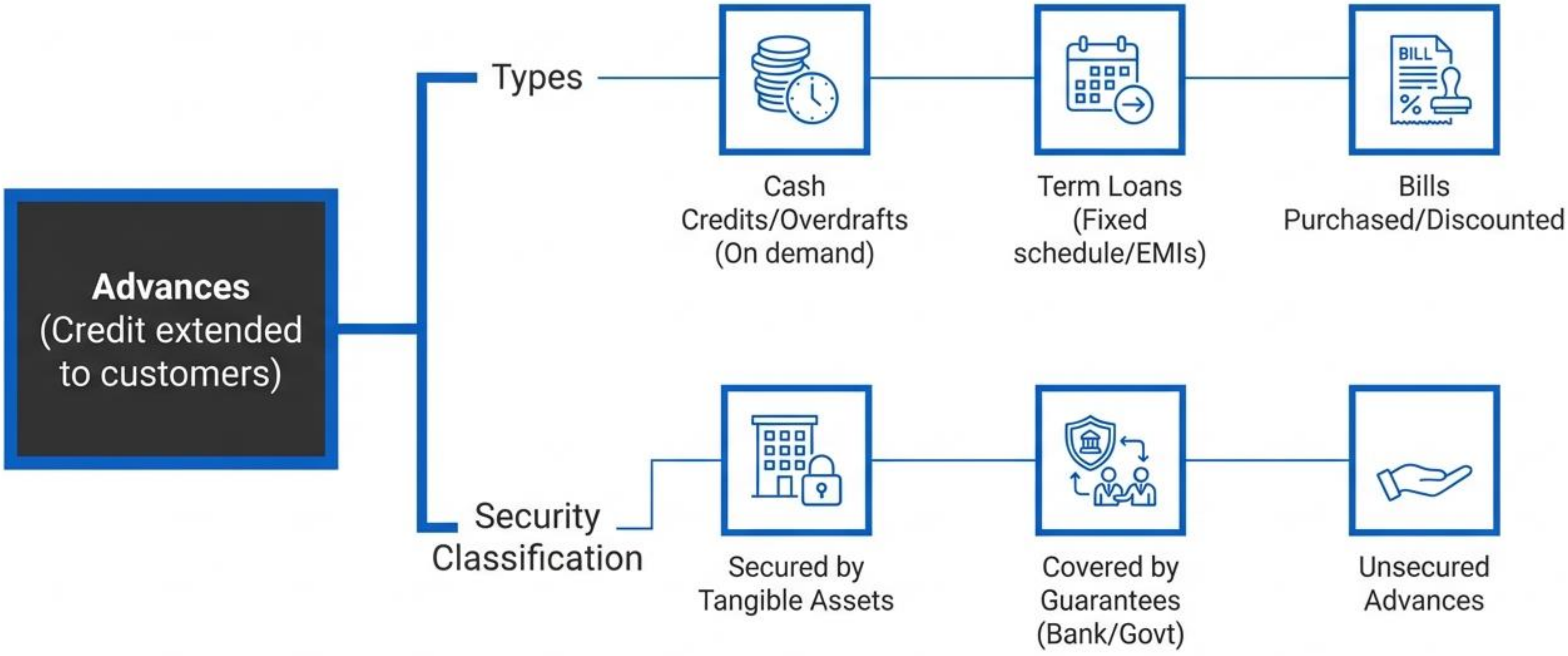
Balances with Banks

Money at Call and Short Notice (Transacted overnight or up to 14 days).

Investments

SLR Compliance (Govt Securities) & Non-SLR (Shares, Debentures, JVs).

Assets: Advances (The Core Business)



Fixed Assets & Other Assets

Fixed Assets

Immovable properties, premises, furniture, hardware, motor vehicles.

Other Assets

- Inter-office Adjustments (Debit balance)
- Interest Accrued (Earned but not collected)
- Tax Paid in Advance / TDS
- Stationery and Stamps
- Non-Banking Assets (Acquired in satisfaction of claims)

FORM B

6 Sch.

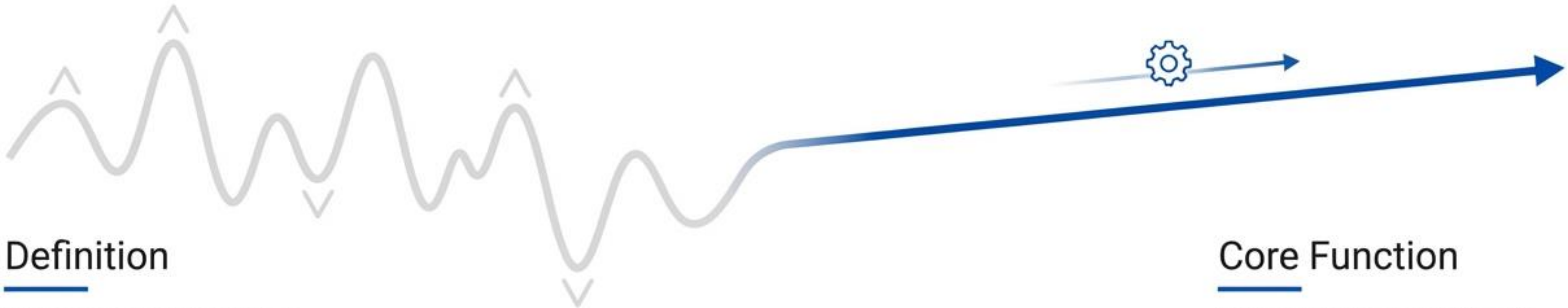
The Scoreboard: Profit and Loss Account

★

INCOME	EXPENSES
13 Interest Income (Advances, Bills, Investments)	15 Interest Expended (Deposits, Borrowings)
14 Other Income (Commission, Exchange, Brokerage, Forex, Profit on sales)	16 Operating Expenses (Staff wages, Rent, Depreciation, Law charges, Insurance)

16 Net P&L A/C

The Physiology: Asset Liability Management (ALM)



Definition

The process of planning, acquiring, and directing the flow of funds to generate adequate stable earnings and build equity.

Core Function

Managing the Net Interest Margin (NIM) to ensure risk/return objectives.

> “ALM is not to avoid risk but to manage risk sustaining profitability.”

Definition and Evolution of ALM:

Banking inherently involves **risk identification, measurement, acceptance, and management** — with **risk management** at its core. Traditionally, banks used administered (fixed/controlled) interest rates to price assets (e.g., loans) and liabilities (e.g., deposits), leading to stable but limited spreads.

- ✓ In a **deregulated, competitive environment** (post-liberalization in India), spreads narrowed due to market-driven rates, competition, and volatility.
- ✓ This shifted focus from partial/isolated management (e.g., credit risk via NPAs, asset-only or liability-only approaches) to **integrated ALM**.
- ✓ ALM evolved from profit-centric views (interest margins) to holistic balance sheet management, considering **all components** (assets, liabilities, maturities, rates, risks).

Core Risks Addressed by ALM:

- ✓ **Interest rate risk** (changes in rates affecting spreads/NIM).
- ✓ **Liquidity risk** (mismatches in cash flows/maturity).
- ✓ **Credit risk** (linked via asset quality).
- ✓ **Market risk** (broader, including forex if applicable).

Why ALM? The Drivers of Significance

Volatility

Deregulation causes interest rate/forex fluctuations, affecting Net Interest Income (NII).

Innovation & Regulation

New products change risk profiles. BIS/RBI mandates ALM policies.

Embedded Options Risk

- Refinancing Risk (Premature loan repayment).
- Reinvestment Risk (Premature deposit withdrawal).

Measuring ALM: Objectives & Parameters

Primary Objectives

Price Matching (Maintaining Spreads)

Price matching maintainting spreads, mean set spread for proper to value price interest.

Liquidity (Maturity Matching)

Liquidity maturity matching to cuva flow rraters no coviosily through and maturity matching

Key Parameters

Net Interest Income (NII)

Interest Income – Interest Expenses

Net Interest Margin (NIM)

Net Interest Income / Average Total Assets

Economic Equity Ratio

Shareholders' Funds / Total Assets

Net Interest Income (NII)

- ✓ NII = Interest Income – Interest Expenses (i.e., total interest earned on assets minus total interest paid on liabilities)
- ✓ **Purpose in ALM**: Measures the **absolute impact** of interest rate volatility (and other mismatches) on **short-term profitability**. Volatility in rates directly affects NII → fluctuations in NII reflect short-term profit instability.
- ✓ **Stabilization Goal**: Minimize fluctuations in NII through effective ALM (e.g., maturity/rate matching, hedging strategies) to keep short-term profits steady.

B L

Net Interest Margin (NIM)

Definition: NIM = (Net Interest Income) / Average Total Assets (expressed as a percentage)

Interpretation: Represents the **spread** (profit margin) earned on the bank's earning assets after covering the cost of funds. It shows how efficiently the bank converts its total assets into net interest profit.

Significance in ALM:

- ✓ NIM is the most widely watched indicator of core banking profitability.
- ✓ Higher NIM = wider spread between earning and paying rates.
- ✓ There is a **direct risk-return trade-off**: Wider spreads often come with higher risk exposure (e.g., lending to riskier borrowers or longer maturities).
- ✓ Banks aim to **maximize NIM** while managing (not eliminating) risks — the key to success lies in **risk management**, not risk avoidance.

Stabilization Goal: Protect and optimize NIM against interest rate changes, product innovations, embedded options, and competitive pressures.

Economic Equity Ratio

Definition: Economic Equity Ratio = Shareholders' Funds (Capital + Reserves & Surplus) / Total Assets (also called Equity to Total Assets ratio or Owned Funds ratio)

Purpose in ALM: Measures the proportion of **owned funds** (equity) versus **borrowed funds** (deposits + borrowings) in the bank's total balance sheet. Tracks shifts in the capital structure over time.

Significance:

- ✓ Indicates the bank's **sustenance capacity** (long-term resilience and ability to absorb losses).
- ✓ Higher ratio = stronger equity buffer → better protection against risks and shocks (aligns with Capital Adequacy Ratio/CAR under Basel norms).
- ✓ Lower ratio = higher leverage → increased vulnerability to losses but potentially higher returns on equity.

Stabilization Goal: Maintain a healthy and stable equity ratio to support long-term earnings quality and financial soundness.

Parameter	Formula	Measures / Focus	Time Horizon	ALM Objective
Net Interest Income (NII)	Interest Income – Interest Expenses	Absolute short-term profit impact of volatility	Short-term	Minimize fluctuations
Net Interest Margin (NIM)	NII / Average Total Assets	Spread efficiency & core profitability	Short- to medium-term	Optimize & protect against rate changes
Economic Equity Ratio	Shareholders' Funds / Total Assets	Capital structure & sustenance capacity	Long-term	Ensure long-term stability & resilience

ALM as Co-ordinated Balance Sheet Management

Stage 1: Specific Management Functions

Asset Side:
- Reserve, Liquidity, Investment, Loan Mgmt.

Liability Side:
- Liability, Reserve Position, Capital Mgmt.

Stage 2: Income-Expense Functions

Profit Formula: Interest Income – Interest Expense – Provisions + Non-interest Revenue – Expenses – Taxes.

Goals: Spread Management, Loan Quality, Capital Adequacy.

Q. Which of the following statements are correct in the context of Asset and Liability Management (ALM) and a bank's balance sheet components?

1. ALM involves only the short-term liquidity management of a bank and is not linked to long-term strategic planning.
2. Asset Liability Committees (ALCOs) in banks have significant authority to influence the institution's asset-liability structure and business alignment.
3. As per the Banking Regulation Act, 1949, commercial banks must prepare their Balance Sheet and Profit & Loss account in the format prescribed in the Third Schedule.
4. In a bank's balance sheet, liabilities and net worth are classified as applications of funds.

SOF

5. Fixed assets, investments, and advances are considered uses of funds on the bank's balance sheet.

A. Only 2, 3, and 5
B. Only 1, 2, and 4
C. Only 3 and 4
D. Only 1, 3, and 5

Answer: A

Solution:

Statement 1 is incorrect: ALM is not limited to short-term liquidity but includes long-term strategic planning, rate sensitivity, and goal setting.

Statement 2 is correct: ALCOs are empowered to shape asset-liability structure and decide discontinuation of activities misaligned with business strategy.

Statement 3 is correct: Under Section 29 of the Banking Regulation Act, 1949, banks must use the format in the Third Schedule as notified by the Government.

Statement 4 is incorrect: Liabilities and net worth are sources of funds, not applications.

Statement 5 is correct: Fixed assets, investments, and advances are uses (applications) of funds.

Q. Which of the following statements regarding the classification of bank expenses is/are correct?

1. Interest paid on participation certificates and penal interest are included under "Interest on Deposits."
2. Advertisement, publicity charges, and their related printing expenses fall under Operating Expenses.
3. Provisions for bad debts and taxation are not considered part of operating expenses but are classified under Provisions and Contingencies.

A) Only statements 1 and 3 are correct
B) Only statements 2 and 3 are correct
C) Only statements 1 and 2 are correct
D) All statements (1, 2, and 3) are correct

Answer: B

Solution:

Statement 1 – Incorrect: Interest on participation certificates and penal interest are categorized under "Others" within Interest Expended, not under "Interest on Deposits."

Statement 2 – Correct: Advertisement and publicity, along with related printing costs, are clearly specified under Operating Expenses.

Statement 3 – Correct: Provisions for bad debts, taxation, etc., are part of “Provisions and Contingencies,” not Operating Expenses.

Q. Which of the following statements regarding the classification of bank expenses is/are correct?

1. Insurance premium paid to DICGC is included in the head "Other Expenditure."
2. Depreciation on leasehold properties and lifts is grouped under Depreciation on Bank's Property.

Choose the correct option:

A. Only Statement 1 is correct
B. Only Statement 2 is correct
C. Both statements are correct
D. Neither 1 nor 2 is correct

Answer: B

Solution:

Statement 1 – Incorrect: Insurance premium paid to DICGC is a part of Insurance, not "Other Expenditure."

Statement 2 – Correct: Depreciation on bank's leasehold properties, lifts, and similar assets is included under Depreciation on Bank's Property.

Q. Which of the following statements regarding Asset Liability Management (ALM) in banks is/are correct?

1. ALM involves only the management of liabilities to ensure banks can meet withdrawal demands and maintain liquidity.
2. ALM focuses on matching the maturity profiles of assets and liabilities to manage interest rate and liquidity risks.
3. ALM includes decisions on the types and mix of both assets and liabilities, aligning them with the bank's profit objectives.

A) Only statements 1 and 3 are correct
B) Only statements 2 and 3 are correct
C) Only statements 1 and 2 are correct
D) All statements (1, 2, and 3) are correct

Answer: B

Solution:

Statement 1 is incorrect: ALM involves both assets and liabilities, not just liabilities. It's about integrated balance sheet management.

Statement 2 is correct: One of the core objectives of ALM is to match asset-liability maturity profiles to manage interest rate and liquidity risks.

Statement 3 is correct: ALM includes planning and managing the types, mix, and volume of financial assets and liabilities from a profitability perspective.

Q. A commercial bank, ABC Bank Ltd., has recently published its annual balance sheet, and you are tasked with reviewing key classifications under RBI norms. Based on the summary below, identify which of the following statements are correct with respect to regulatory classification of liabilities and assets:

ABC Bank Ltd. has reported the following:

- ₹1,200 crore in share capital and reserves, comprising statutory reserves and retained earnings.
- ₹35,000 crore in deposits, including ₹2,000 crore in overdue term deposits.
- ₹1,800 crore borrowed from NABARD for rural refinance and ₹500 crore from RBI under LAF.
- ₹4,000 crore shown as "bills payable" including bankers' cheques and telegraphic transfers.
- ₹20,000 crore invested in SLR-eligible government securities and ₹2,000 crore in listed debentures.
- ₹28,000 crore extended as advances, of which ₹8,000 crore are secured by tangible assets, ₹2,000 crore are covered under government guarantees, and the rest are unsecured.

Which of the following statements is/are correct?

1. Share capital and reserves fall under capital and surplus, which are long-term sources of funds forming part of liabilities.
2. Borrowings from NABARD and RBI are treated as liabilities and not deposits.
3. Advances secured by bank/government guarantees are to be reported under secured advances.
4. ~~All investments made in debentures are part of the bank's "Advances" head in the balance sheet.~~
5. ~~Overdue term deposits must be excluded from total deposits in the liabilities section.~~

A. Only 2, 3 and 4 are correct
B. Only 1, 2 and 5 are correct
C. Only 1, 2 and 3 are correct
D. Only 1, 3 and 5 are correct

Answer C

Explanation:

Statement 1 – Correct: Capital and reserves (including statutory reserves and retained earnings) fall under long-term sources of funds and are listed under liabilities.

Statement 2 – Correct: Borrowings from NABARD and RBI are not deposits; they are classified under "Borrowings" on the liability side.

Statement 3 – Correct: Advances covered by government or bank guarantees are classified as "secured advances" under regulatory definitions.

Statement 4 – Incorrect: Investments in debentures are part of the "Investments" head, not "Advances."

Statement 5 – Incorrect: Overdue term deposits are still classified under deposits; they are not excluded from the liability section.

Q. A scheduled commercial bank in India is undergoing a strategic audit of its Asset Liability Management (ALM) practices. The internal audit team flags the following issues during their review for FY 2024–25:

- Nearly 48% of liabilities are maturing within the next 9 months, while only 14% of assets mature in the same period.
- The bank had recently introduced 'Flexi-Deposit' products aggressively, allowing customers to auto-sweep idle balances from savings to term deposits and vice versa.
- An unexpected 25 bps cut in interest rates by RBI has led to a decline in returns from fixed-rate loan portfolios, while the cost of some floating rate deposits has not reduced proportionately.
- The audit also found that nearly 25% of borrowers have exercised prepayment options, while 15% of depositors opted for premature withdrawals.
- Despite the above, the bank's management continues to focus on credit growth and retail deposit mobilization, without a dedicated ALM monitoring committee.

Which of the following risks or concerns must be urgently addressed through a structured ALM framework in light of the above findings?

1. Liquidity Risk due to maturity mismatches between assets and liabilities
2. Reinvestment and Refinancing Risk arising from embedded options exercised by customers
3. Interest Rate Risk affecting Net Interest Margin (NIM) due to market rate volatility

A) Only statements 1 and 3
B) Only statements 2 and 3
C) Only statements 1 and 2
D) All statements (1, 2, and 3)

Answer D

Solution:

Statement 1: Valid — Significant maturity mismatch directly triggers liquidity risk.

Statement 2: Valid — Customer behavior (prepayments and premature withdrawals) exposes the bank to reinvestment and refinancing risk due to embedded options.

Statement 3: Valid — Rate cut impacts asset returns while liability costs remain sticky, affecting Net Interest Margin, hence interest rate risk.

Q. Which of the following statements are correct?

1. Asset Liability Management focuses solely on matching the maturity profile of assets and liabilities to avoid liquidity risk.
2. A key objective of ALM at the macro level is to support product pricing strategies and capital allocation.
3. Net Interest Margin (NIM) is calculated by subtracting interest expenses from interest income.

$\frac{NII}{TA}$

A) Only statements 1 and 3 are correct
B) Only statements 2 and 3 are correct
C) Only statements 2 is correct
D) All statements (1, 2, and 3) are correct

Answer: C

Solution:

Statement 1 – Incorrect: ALM does manage liquidity risk, but not solely. It also deals with interest rate risk, credit risk, forex risk, etc.

Statement 2 – Correct: At the macro level, ALM helps in capital allocation, policy formulation, and product pricing.

Statement 3 – Incorrect: This is the formula for Net Interest Income (NII), not NIM. NIM is NII divided by average total assets.

Q. ABC Bank Ltd. is preparing its Profit & Loss Account for the financial year. The following are the financial activities recorded during the year. As a banking analyst, determine under which head each of these income items will be classified in the Profit and Loss Account of the bank:

- ₹50 crore received as interest on term loans and overdraft facilities.
- ₹5 crore earned as brokerage on sale of government securities and locker rent.
- ₹2 crore received from money market placements and interest on RBI balances.
- ₹4 crore profit on revaluation of HTM (Held to Maturity) investment securities.
- ₹1 crore as recovery from a defaulting customer toward insurance and godown rent charges.

Which of the following combinations correctly categorizes the entries in the appropriate heads of the bank's Profit & Loss Account?

(1) – Interest Income: Interest/Discount on Advances
(2) – Other Income: Commission, Exchange and Brokerage
(3) – Interest Income: Interest on Balances with RBI and Other Interbank Funds
(4) – Other Income: Profit on Revaluation of Investments
(5) – Other Income: Miscellaneous Income

13 + 14

11 01

A. Only 1, 2, 3, and 5 are correct
B. Only 1, 3, 4, and 5 are correct
C. Only 2, 3, 4, and 5 are correct
D. All 1, 2, 3, 4, and 5 are correct

Answer: D

Explanation:

Interest/Discount on Advances includes interest earned on loans like term loans, overdrafts — correctly classifies item (1).

Commission, Exchange and Brokerage includes brokerage on securities and locker rent — covers item (2).

Interest on RBI/Interbank Funds includes interest on RBI balances and money market placements — matches item (3).

Profit on Revaluation of Investments includes net gains or losses on revaluation of securities — applies to item (4).

Miscellaneous Income includes recoveries such as godown rents and insurance — exactly item (5).

Q. Which of the following statements regarding **the classification of bank expenses is/are correct?**

1. Interest paid on participation certificates and penal interest are included under "Interest on Deposits."
2. Advertisement, publicity charges, and their related printing expenses fall under Operating Expenses.
3. Provisions for bad debts and taxation are not considered part of operating expenses but are classified under Provisions and Contingencies.
4. Insurance premium paid to DICGC is included in the head "Other Expenditure."
5. Depreciation on leasehold properties and lifts is grouped under Depreciation on Bank's Property.

Choose the correct option:

A. Only 2, 3 and 5
B. Only 1, 2 and 4
C. Only 2, 3, 4 and 5
D. Only 3 and 5

Answer: A

Solution:

Statement 1 – Incorrect: Interest on participation certificates and penal interest are categorized under "Others" within Interest Expended, not under "Interest on Deposits."

Statement 2 – Correct: Advertisement and publicity, along with related printing costs, are clearly specified under Operating Expenses.

Statement 3 – Correct: Provisions for bad debts, taxation, etc., are part of “Provisions and Contingencies,” not Operating Expenses.

Statement 4 – Incorrect: Insurance premium paid to DICGC is a part of Insurance, not “Other Expenditure.”

Statement 5 – Correct: Depreciation on bank’s leasehold properties, lifts, and similar assets is included under Depreciation on Bank’s Property.

Q. Which of the following statements regarding Asset Liability Management (ALM) in banks is/are correct?

1. ALM involves only the management of liabilities to ensure banks can meet withdrawal demands and maintain liquidity.
2. ALM focuses on matching the maturity profiles of assets and liabilities to manage interest rate and liquidity risks.
3. ALM includes decisions on the types and mix of both assets and liabilities, aligning them with the bank's profit objectives.
4. ALM seeks to completely avoid all risks in the banking business to ensure capital safety and regulatory compliance.
5. ALM requires dynamic restructuring of the balance sheet portfolio in response to changes in economic conditions and interest rate expectations.

A. 1 and 4 only
B. 1, 3, and 4 only
C. 2, 3, and 5 only
D. 2, 4, and 5 only

Answer: C

Solution:

Statement 1 is incorrect: ALM involves both assets and liabilities, not just liabilities. It’s about integrated balance sheet management.

Statement 2 is correct: One of the core objectives of ALM is to match asset-liability maturity profiles to manage interest rate and liquidity risks.

Statement 3 is correct: ALM includes planning and managing the types, mix, and volume of financial assets and liabilities from a profitability perspective.

Statement 4 is incorrect: The goal of ALM is not to avoid risk entirely but to manage risk strategically while sustaining profitability.

Statement 5 is correct: ALM includes dynamic balance sheet restructuring to respond to evolving economic and interest rate scenarios.

Q. You are part of the ALCO (Asset Liability Committee) in a scheduled commercial bank reviewing the bank's current ALM strategy. The following observations are made during the quarterly review:

- The bank's liquidity buffers are falling below internal thresholds due to aggressive loan disbursals.
- A large chunk of investments is locked in low-yield long-term securities while short-term liabilities are mounting.
- Interest income has increased due to higher loan disbursement, but net profit is stagnating due to rising non-interest expenses and provisioning.
- The bank's service fee income remains flat across quarters.
- Tier-1 capital adequacy has dropped marginally following increased risk-weighted assets from recent corporate loan exposures.

Which of the following corrective actions would best align with the objectives of coordinated balance sheet management under ALM framework?

1. Improve spread management by optimizing interest margins through re-pricing of loans and liabilities.
2. Address investment-security mismatch by restructuring the investment portfolio to align asset-liability tenors.
3. Generate non-interest revenue by revising service charges and introducing new fee-based products.
4. Initiate cost-cutting measures focused on non-interest operating expenses to improve profit margins.
5. Infuse fresh equity capital and reduce exposure to high-risk corporate loans to restore capital adequacy ratios.

A. Only 1, 2 and 4
B. Only 1, 3, 4 and 5
C. Only 2, 3 and 5
D. All of the above

Answer: D

Solution:

Each of the five suggested actions addresses a specific facet of the ALM framework:

1: Spread management tackles the core income-expense aspect of Stage 2.

2: Realigning the investment portfolio handles asset-side liquidity and tenor risk—Stage 1.

3: Boosting fee income directly supports one of the listed ALM objectives—Stage 2.

4: Cutting non-interest expenses aligns with income-expense optimization—Stage 2.

5: Strengthening capital adequacy is a critical long-term balance sheet management function—Stage 1.

26.5.1 Net Interest Income (NII)

The impact of volatility on the short-term profit is measured by Net Interest Income.

Net Interest Income = Interest Income – Interest Expenses.

In order to stabilise short-term profits; banks have to minimise fluctuations in the NII.

26.5.2 Net Interest Margin (NIM)

Net Interest Margin is defined as net interest income divided by average total assets.

Net Interest Margin (NIM) = Net Interest Income/Average total Assets.

Net Interest Margin can be viewed as the 'Spread' on earning assets.

The net income of banks comes mostly from the spreads maintained between total interest income and total interest expense. The higher the spread, the more will be the NIM. There exists a direct correlation between risks and return. As a result, greater spreads only imply enhanced risk exposure. But since any business is conducted with the objective of making profits and achieving higher profitability is the target, it is the management of risks and not risk elimination, that holds the key to success.

Q. A commercial bank provides the following information for the financial year 2025–26 (₹ in crore):

Particulars	Average Amount	Average Yield/Cost
Cash Credit & Working Capital Loans	22,750	11.20%
Retail Mortgage Loans	14,600	8.95%
Investments in Government Securities	11,250	7.42%
Call Money Lending	4,850	6.35%
Savings Bank Deposits	16,400	3.10%
Current Deposits	8,750	0%
Bulk Term Deposits	18,900	7.55%
Refinance & Interbank Borrowings	6,300	6.85%

2548
1306.70
834.75
307.90
508.4
1426.95
431.55
2366.9

4997.35
−96

3.70%

Additional Information:

- Interest income of ₹96 crore on certain stressed assets remained unrealised and was reversed during the year.
- Processing fees and treasury profits are not to be considered.
- Average total assets of the bank during the year were ₹68,500 crore.
- Interest on current deposits is nil.

2534.45
= 3.70

The bank wants to compute its Net Interest Margin (NIM). What is the Net Interest Margin (approx.) of the bank?

A. 2.41%

B. 2.67%

C. 2.94%

D. 3.18%

3.70%

Answer: D

Solution:

Step 1: Compute Interest Income

Cash Credit & WC Loans = 22,750 × 11.20% = ₹2,548 crore; Retail Mortgage Loans = 14,600 × 8.95% = ₹1,306.70 crore; Government Securities = 11,250 × 7.42% = ₹834.75 crore; Call Money Lending = 4,850 × 6.35% = ₹307.975 crore

Gross Interest Income = 2,548 + 1,306.70 + 834.75 + 307.975 = ₹4,997.425 crore

Less: Unrealised Interest Reversed = ₹96 crore

Adjusted Interest Income = ₹4,901.425 crore

Step 2: Compute Interest Expense

Savings Deposits = 16,400 × 3.10% = ₹508.40 crore; Current Deposits = Nil; Bulk Term Deposits = 18,900 × 7.55% = ₹1,426.95 crore; Refinance & Borrowings = 6,300 × 6.85% = ₹431.55 crore

Total Interest Expense = 508.40 + 1,426.95 + 431.55 = ₹2,366.90 crore

Step 3: Compute Net Interest Income (NII): NII=4901.425−2366.90 = ₹2,534.525 crore

Step 4: Compute Net Interest Margin (NIM): NIM= 2534.525/ 68500 = 0.036999 ≈ 3.70%

UNIT 27

Capital Adequacy – Basel Norms

STRUCTURE

Blueprint of the Basel Framework

The Framework: The Three Pillars of Basel III Overview

Pillar 1:
Minimum Capital Requirements

- CRAR
- Tier-1/Tier-2 Constituents
- Risk Assessment Approaches

Pillar 2:
Supervisory Review Process & ICAAP

Bank & Supervisor Responsibilities, Methodology, Outcomes

Pillar 3:
Market Discipline

- Guiding Principles
- Disclosure Scope
- Tables DF-1 to DF-10

Foundation: Introduction & Scope of Application (Applicability Rules & Objectives)

The **Revised Basel Framework** (primarily **Basel II**) is built on **three mutually reinforcing pillars** to ensure banks maintain adequate capital and manage risks effectively.

Pillar 1: Minimum Capital Requirements Focuses on calculating capital needed for key risks, offering **risk-sensitive approaches** (more advanced = more sophisticated):

- ✓ **Credit risk**: Standardized Approach, Foundation IRB, Advanced IRB.
- ✓ **Market risk**: Standardized (maturity ladder/duration-based) or Advanced (e.g., internal models like VaR).
- ✓ **Operational risk**: Basic Indicator Approach, Standardized Approach, Advanced Measurement Approach.

Banks select approaches suited to their operations and development stage.

Pillar 2: Supervisory Review Process (SRP) / ICAAP Ensures banks hold enough capital for **all risks** (not just Pillar 1). Supervisors review banks' **Internal Capital Adequacy Assessment Process (ICAAP)** to:

- ✓ Cover risks **not fully captured** in Pillar 1.
- ✓ Address risks **not included** in Pillar 1.
- ✓ Consider **external factors**. It also promotes better risk management techniques.

Pillar 3: Market Discipline Requires banks to disclose key information on:

•Scope of application, capital, risk exposures, risk assessment processes, and capital adequacy.

This enables market participants (investors, etc.) to assess the bank's risks and make informed decisions, complementing Pillars 1 and 2.

Scope of Application in India

- ✓ The revised capital adequacy norms apply uniformly to **all commercial banks** (excluding **co-operative banks**, **Local Area Banks**, and **Regional Rural Banks**).
- ✓ They are implemented at the **consolidated/group level** (where a licensed bank is the controlling entity), covering all group entities under its control.
- ✓ Consolidated banks include banking and financial subsidiaries but typically **exclude** entities in insurance or non-financial services businesses.
- ✓ Banks must maintain a minimum **Capital to Risk-Weighted Assets Ratio (CRAR)** on an ongoing basis.
- ✓ For **operational risk**, all applicable banks (excluding exempted ones) must use the **Standardised Approach (SA)**.

Regional Rural Banks and others follow simpler approaches (e.g., Basic Indicator Approach for operational risk, Standardised Duration Approach for market risk).

National regulators (RBI) can set higher minimum capital standards than Basel minima and promote forward-looking, evolving standards.

Key Objectives and Features of Basel III

- ✓ Builds on the 1988 Accord to strengthen the soundness and stability of the banking system.
- ✓ Promotes stronger **risk management** practices and better risk assessment.
- ✓ Offers a range of options for calculating capital requirements for **credit risk** and **operational risk** (with increasing sophistication).
- ✓ Encourages banks to invest in sophisticated internal risk management systems.
- ✓ Provides incentives for better risk management through disclosures and processes.
- ✓ Seen as an opportunity for banks to demonstrate strong risk credentials, though it may initially increase administrative burden.
- ✓ Emphasizes cooperation between home and host country supervisors.

Three Pillars of Basel III (restated briefly) The framework rests on the same three pillars as Basel II, with enhancements:

Pillar 1: Minimum capital requirements for credit, market, and operational risks.

Pillar 2: Supervisory review process to ensure comprehensive risk coverage.

Pillar 3: Market discipline through enhanced disclosures.

Establishing the Scope of Application

Applicability:

- The revised capital adequacy norms are applicable uniformly to all Commercial Banks, both at the solo level (global position) and the consolidated level.

Exceptions:

- Specifically excludes Co-operative Banks, Local Area Banks, and Regional Rural Banks (RRBs).

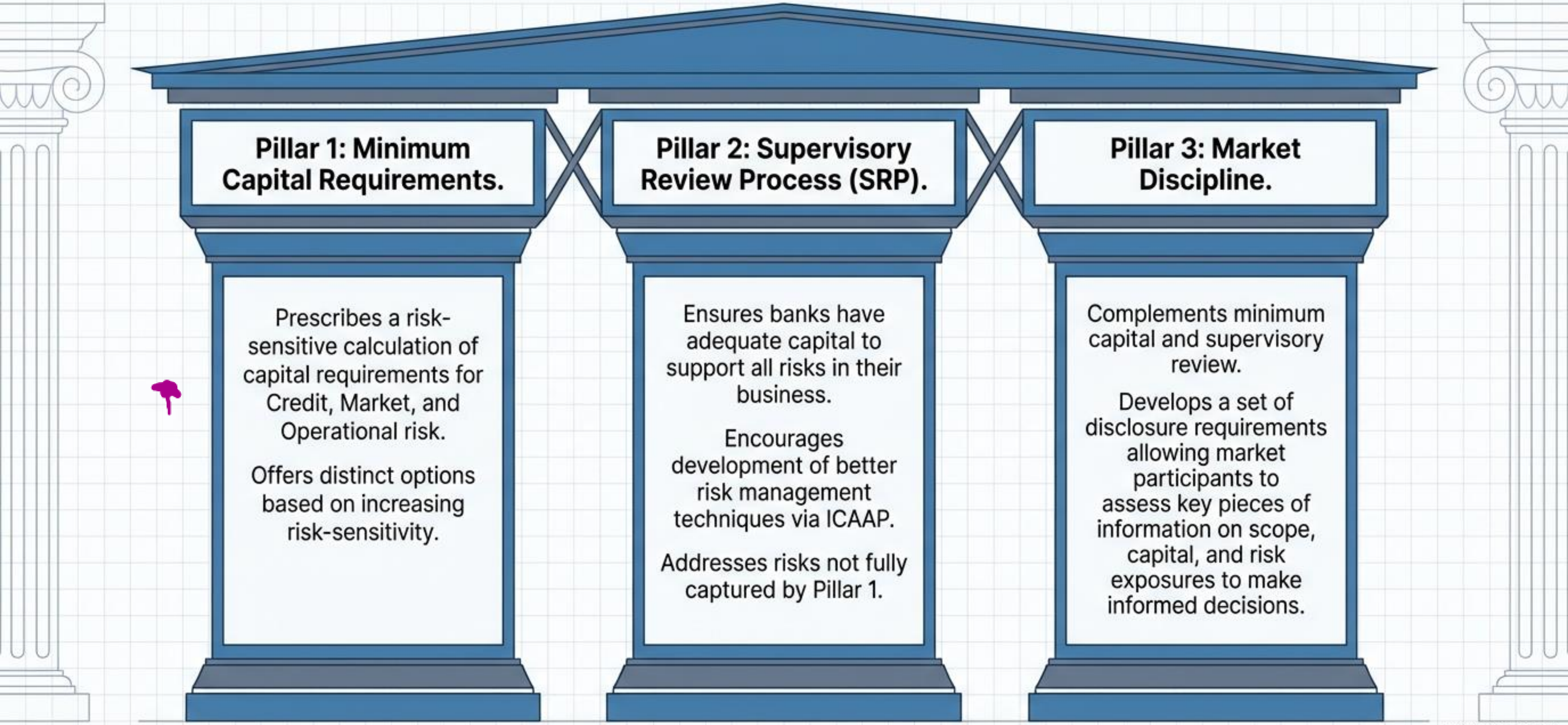
Consolidation Rules:

- A consolidated bank includes all group entities under its control, except exempted entities.
- A consolidated bank may exclude group companies engaged in insurance business and businesses not pertaining to financial services.

Core Mandate:

- A consolidated bank must maintain a minimum Capital to Risk-weighted Assets Ratio (CRAR) as applicable to a bank on an ongoing basis.
- The fundamental objective is to strengthen the soundness and stability of the banking system by promoting stronger risk management practices.

The Three Mutually Reinforcing Pillars of Basel III



Pillar 1 Defines Strict Minimum Capital Requirements

The CRAR Calculation

Capital ratio is calculated using the definition of regulatory capital and risk-weighted assets (RWA).
Total RWA = Risk weighted assets for credit risk + 12.5 * Capital requirement for market risk + 12.5 * Capital requirement for operational risk.

Minimum Threshold: The total capital ratio must not be lower than 9% in India (Basel III minimum is 8%).

Tier-1 Capital (Core Capital)

Consists of paid-up capital, free reserves, and unallocated surpluses, less specified deductions. Any capital requirement arising in respect of credit and counter-party risk needs to be met by Tier-1 and Tier-11 capital.

Tier-2 Capital (Supplementary Capital)

Comprises subordinated debt of more than five years' maturity, loan loss reserves, revaluation reserves, investment fluctuation reserves, and limited life preference shares. Tier-II capital is restricted to 100% of Tier-I capital. (Note: Tier-III has been phased out under Basel III).

Pillar 1: Minimum Capital Requirements (under Basel III, as implemented in India by RBI)

Capital Definition and Components Regulatory capital is calculated using definitions largely carried over from earlier accords but enhanced under Basel III. It includes:

- ✓ **Tier 1 Capital** (going-concern capital): Primarily **Common Equity Tier 1 (CET1)** (e.g., paid-up equity, free reserves, unallocated surpluses, retained earnings) + **Additional Tier 1** instruments.
- ✓ **Tier 2 Capital** (gone-concern/supplementary capital): Subordinated debt (>5 years maturity), loan loss reserves, limited preference shares, revaluation reserves, etc. (Tier 3 capital for market risk, introduced in Basel II, has been phased out in Basel III.)

Deductions apply for certain items (e.g., goodwill, specified investments).

Minimum Capital Ratios (India-specific, more conservative than global Basel III minima)

Banks must maintain a minimum **Capital to Risk-Weighted Assets Ratio (CRAR)** of **9%** (higher than Basel's 8%).

In practice, with **Capital Conservation Buffer (CCB)** of 2.5% (in CET1), the effective minimum is **11.5%**.

Breakdown (full implementation):

- ✓ CET1 ≥ **5.5%** of RWAs (+ CCB 2.5% → 8.0%).
- ✓ Tier 1 ≥ **7%** of RWAs.
- ✓ Total Capital (Tier 1 + Tier 2) ≥ **9%** of RWAs (+ CCB → 11.5%).

Methodologies for Computing Capital Requirements

Credit Risk Approaches

- Standardised Approach
- Internal Rating Based (IRB) Approaches (Foundation Approach; Advanced Approach)

Market Risk Approaches

- Standardised method (Maturity method; Duration method)
- Internal models method

Operational Risk Approaches

- Basic Indicator Approach (BIA)
- Standardised Approach
- Advanced Measurement Approach (AMA)

Exam Note: RBI decided that commercial banks in India (excluding LABs and RRBs) shall adopt the Standardised Approach for credit risk and Basic Indicator Approach for operational risk, while applying Standardised Duration Approach for market risk.

Risk-Weighted Assets (RWA) Calculation Total RWAs = Credit Risk RWAs + (Market Risk capital requirement × 12.5) + (Operational Risk capital requirement × 12.5). (The 12.5 multiplier is the reciprocal of 8%, converting capital charge to equivalent RWAs.)

- ✓ **Credit Risk**: Calculated using risk weights on assets/exposures (Standardised Approach common in India; IRB for advanced banks).
- ✓ **Market Risk**: Standardized (e.g., duration-based) or advanced (internal models).
- ✓ **Operational Risk**: Basic Indicator, Standardized, or Advanced Measurement Approach (phased toward standardized in recent updates).

This makes the framework more **risk-sensitive** than Basel I/II, incentivizing better internal risk management.

Key Enhancements in Basel III (Pillar 1)

Stronger focus on quality of capital (more CET1).

Better capture of all risks (credit, market, operational).

Supervisory review (Pillar 2) and market discipline (Pillar 3) complement Pillar 1.

The text includes a diagram showing the **three pillars** of Basel III:

- ✓ **Pillar I** — Enhanced Minimum Capital & Liquidity Requirements
- ✓ **Pillar II** — Enhanced Supervisory Review Process for Time-wise Risk Management & Capital Planning
- ✓ **Pillar III** — Enhanced Risk Disclosures & Market Discipline

Supervisory Review Process (SRP) and The Four Key Principles

Core Objective: Pillar 2 addresses risks not fully captured by Pillar 1 (e.g., credit concentration), risks not taken into account at all (e.g., liquidity, strategic, reputational), and factors external to the bank.

Banks' Responsibilities

- (Principle 1): Banks should have a process for assessing their overall capital adequacy in relation to their risk profile and a strategy for maintaining capital levels.
- (Principle 3): Banks should operate above the minimum regulatory capital ratios.

Supervisors' Responsibilities

- (Principle 2): Supervisors should review and evaluate a bank's ICAAP and take appropriate action if not satisfied. Review compliance with regulatory ratios.
- (Principle 3): Supervisors should have the ability to require banks to hold capital in excess of the minimum.
- (Principle 4): Supervisors should seek to intervene at an early stage to prevent capital falling below minimums, and require rapid remedial action if not maintained.

Structural Aspects of the ICAAP

Defining ICAAP: Comprises procedures and measures designed to ensure appropriate identification and measurement of risks, an appropriate level of internal capital in relation to the risk profile, and application/further development of risk management systems.

Board and Senior Management Oversight: The ultimate responsibility for designing and implementing ICAAP lies with the Board of Directors. They must approve the design to ensure it forms an integral part of the management process and decision-making culture.

Firm-wide Application: Every bank must have an ICAAP prepared on a solo basis (every tier) and at the consolidated level. Senior management must integrate this into firm-wide risk oversight to identify emerging risks timely.

Review Process: The Board must assess and document, at least once a year, whether processes successfully achieve the envisaged objectives.

Methodologies Powering the ICAAP

A Forward-Looking Process: Must account for expected future developments (strategic plans, macro-economic factors) and future constraints in capital availability. Requires an explicit, Board-approved capital plan detailing capital needs, utilization, desired levels, and contingency plans.

A Risk-Based Process: Capital targets must be consistent with the risk profile. For readily quantifiable risks, a quantitative approach forms the foundation. For less quantifiable risks (reputation, strategic), the focus is on qualitative assessment, risk management, and mitigation.

Stress Tests and Scenario Analyses: Management must conduct periodic stress tests for material risk exposures to evaluate vulnerabilities to unlikely but plausible extreme market conditions.

Risk Aggregation: Banks must assess risks across the entire bank, understanding challenges in aggregation and documenting diversification assumptions rigorously.

ICAAP as a Forward-looking Process

- ICAAP must be **forward-looking**, considering future developments such as:
    - Strategic and business plans
    - Macro-economic conditions
    - Future constraints on availability and use of capital
- Banks must maintain **adequate capital** in line with the **nature, scale, complexity, and risk profile** of both on-balance-sheet and off-balance-sheet activities.
- Capital strategy should clearly show how it aligns with **macro-economic factors**.
- Banks must have a **Board-approved capital plan** specifying:
    - Capital objectives and target levels
    - Time horizon to achieve these targets
    - Capital planning process and assigned responsibilities
- The capital plan should cover:
    - Capital needs
    - Expected capital utilisation
    - Desired capital level
    - Capital-related limits
    - Contingency plans for unexpected events and deviations

ICAAP as a Risk-based Process

- Capital adequacy depends on the **bank's risk profile and operating environment**.
- Capital targets must be consistent with the **level and types of risks** undertaken.
- ICAAP should cover **all material risks** faced by the bank.
- For risks that are **difficult to quantify** (e.g. strategic and reputational risks):
    - Greater emphasis should be on **qualitative assessment**, risk management, and mitigation.
- The ICAAP document must clearly distinguish:

ICAAP to Include Stress Tests and Scenario Analyses (Pillar 2 – Key Forward-Looking Requirement)

As part of the **Internal Capital Adequacy Assessment Process (ICAAP)**, banks (under RBI guidelines aligned with Basel) must incorporate **stress testing** and **scenario analysis** as a mandatory, minimum requirement. This ensures a forward-looking evaluation of capital resilience.

Core Requirement Bank management shall, **at a minimum**, periodically conduct **relevant stress tests**, focusing on the bank's **material risk exposures**. The goal is to evaluate potential **vulnerability** to **unlikely but plausible** adverse events or movements in market/economic conditions that could severely impact the bank.

Purpose and Benefits

- ✓ Assess how extreme circumstances (e.g., severe economic downturns, market shocks, or other tail events) could adversely affect capital, earnings, liquidity, or overall financial position.
- ✓ Provide better understanding of the bank's likely exposure in **extreme but plausible** scenarios.
- ✓ Help management identify vulnerabilities, plan capital buffers, and strengthen risk management.

Scope of Risks Covered Stress testing should rigorously address **material risks**, including (but not limited to):

- ✓ Credit risk
- ✓ Market risk
- ✓ Operational risk
- ✓ Interest Rate Risk in the Banking Book (**IRRBB**)
- ✓ Credit concentration risk
- ✓ Liquidity risk
- ✓ Other risks (e.g., emerging risks like reputational or strategic, as applicable)

Operational Aspects This section outlines the broader **risk universe** expected to be captured in ICAAP (beyond Pillar 1 minima):

Identifying and Measuring Material Risks: The primary objective of ICAAP is comprehensive identification of all material risks.

- ✓ Quantifiable risks should use robust data-supported methods (e.g., models, historical data, statistical techniques).
- ✓ Measurement approaches vary by bank but must be sound and proportionate.

Stress testing complements regular risk assessment by simulating severe conditions, often through:

Scenario analysis (e.g., historical crises, hypothetical macro shocks).

Sensitivity analysis (isolated variable shocks).

Integration into capital planning (e.g., forward-looking capital needs under stress).

In RBI practice (from Master Circular on Basel III and related guidelines):

ICAAP must be **forward-looking** and **risk-based**.

Stress tests/scenario analyses are explicitly required to inform internal capital targets (often above regulatory minima).

Results feed into supervisory review (SREP), where RBI may impose additional capital if vulnerabilities are identified.

Quantitative and Qualitative Approaches in ICAAP

•Risk measurement should combine **both quantitative and qualitative elements**.

•Wherever possible, **quantitative methods should form the foundation** of risk measurement.

•Higher **model uncertainty or business complexity** should lead to a **larger capital cushion**.

•Quantitative models based only on **most-likely outcomes** (used for budgeting/forecasting) are **insufficient for capital adequacy**, as ICAAP must also consider **low-probability, high-impact events**.

•**Stress testing and scenario analysis** are essential tools to assess the impact of extreme but plausible adverse events on bank safety and soundness.

Risk Aggregation and Diversification Effects

•ICAAP should assess risks **across the entire bank**, not in silos.

•Banks aggregating risks across risk types or business lines must understand and manage the **challenges of aggregation**.

•While recognizing diversification benefits, management must:

- Be **systematic and rigorous** in documenting decisions
- Clearly state **assumptions** used at each level of aggregation

•Banks should have **robust systems** for risk aggregation, considering:

- Data quality and consistency
- Correlation assumptions across risks
- Changes in correlations over time, especially under **stressed market conditions**

Reviewing Key ICAAP Outcomes and Document Structure

- **The ICAAP Document:** A comprehensive paper detailing the ongoing assessment of the entire risk spectrum, mitigation strategies, and capital requirements. Must explicitly cover methodologies, assumptions, and validation work for internal models.
- **Capital Transferability:** Must detail any contractual, commercial, regulatory, or statutory restrictions on the management's ability to transfer capital or distribute dividends across conglomerate structures.
- **Assessment of Specific Risks:** Must include explicit assessments for Off-balance Sheet Exposures (focusing on securitisation and potential maturity mismatches), Reputational Risk (implicit support provisions), Valuation and Liquidity Risk, and Stress Testing Practices.
- **The Principle of Proportionality:** RBI expects the degree of sophistication adopted in the ICAAP for risk measurement and management to be commensurate with the nature, scope, scale, and complexity of the bank's business operations.

What is an ICAAP document?

A **comprehensive document** explaining:

- ✓ The bank's **ongoing assessment of all material risks**
- ✓ How these risks are **measured and mitigated**
- ✓ How much **capital is required and maintained** in relation to the risk profile

Risks Analysed under ICAAP

The document should identify and analyse major risks faced by the bank, including:

- ✓ Credit risk
- ✓ Market risk
- ✓ Operational risk
- ✓ Liquidity risk
- ✓ Concentration risk
- ✓ Interest rate risk in the banking book (IRRBB)
- ✓ Residual risk of securitisation
- ✓ Strategic risk
- ✓ Business risk
- ✓ Reputational risk
- ✓ Pension obligation risk
- ✓ Other residual risks
- ✓ Any additional risks identified by the bank

Assessment & Reporting Requirements

- ✓ For each identified risk, the ICAAP document should include:
- ✓ Explanation of **how the risk is assessed**, and wherever possible, **quantitative results**
- ✓ Description of **risk mitigation measures**, especially for risks highlighted during **Reserve Bank of India on-site inspections**
- ✓ **Comparison of CRAR** assessed by RBI during inspection with CRAR computed under ICAAP
- ✓ Clear articulation of the bank's **risk appetite**, expressed quantitatively by risk category
- ✓ Explanation of the **consistency ("fit")** between risk appetite and overall risk assessment
- ✓ Details of **non-capital risk mitigation methods** (policies, controls, limits, etc.), where applicable

Pillar 3 Demands Strict Market Discipline

Five Guiding Principles for Disclosures:

1. Clear
2. Comprehensive
3. Meaningful to users
4. Consistent over time
5. Comparable across banks.

The Purpose

To complement minimum capital requirements and the supervisory review process by developing a set of disclosure requirements that allow market participants to assess key pieces of information regarding capital adequacy.

Scope and Frequency

Applies at the top consolidated level. Disclosures must be made at least on a half-yearly basis, but specific elements (Capital Adequacy, General Credit Risk, Portfolios under Standardised Approach) must be made quarterly on the bank's website alongside financial results.

Materiality & Proprietary Info

Information is material if omission influences economic decisions. Proprietary information that undermines competitive position may be withheld, but general disclosure principles must still be formally approved by the Board.

Guiding Principles for Pillar 3 Disclosures The Basel Committee on Banking Supervision (BCBS) has established **five core guiding principles** for effective disclosures (adopted in India by RBI):

1. **Clear** — Disclosures should be presented in an understandable manner, avoiding jargon where possible and using plain language.
2. **Comprehensive** — Cover all material aspects of the bank's risk exposures, risk assessment processes, capital adequacy, and risk management practices.
3. **Meaningful to users** — Provide information that is relevant and useful to market participants in assessing the bank's risk profile and soundness.
4. **Consistent over time** — Use consistent formats, definitions, and methodologies across reporting periods to enable trend analysis.
5. **Comparable across banks** — Structured in a way that allows meaningful comparisons between different institutions.

Implementation in India (RBI Context)

- RBI mandates Pillar 3 disclosures for all scheduled commercial banks (with varying detail based on size and complexity).
- Disclosures are typically published quarterly/annually on the bank's website and often in annual reports.

- They include tables/templates on:
    - ✓ Capital structure and adequacy (CRAR, Tier 1, CET1, buffers).
    - ✓ Credit risk exposures, risk weights, NPA details.
    - ✓ Market risk and operational risk approaches.
    - ✓ IRRBB, liquidity coverage, leverage ratio.
    - ✓ Remuneration, governance, and qualitative risk management descriptions.

Pillar 3: Market Discipline – Scope, Frequency, and Validation (Basel III as per RBI Guidelines in India)

Pillar 3 disclosures promote **transparency** and **market discipline** by requiring banks to publicly share detailed information on capital adequacy, risk exposures, and risk management practices. This complements Pillars 1 and 2.

Scope of Application

- ✓ Disclosures apply primarily at the **top consolidated level** of the banking group (where the parent/holding bank is the top consolidated entity).
- ✓ This covers the entire banking group under the Capital Adequacy Framework.
- ✓ Disclosures related to individual subsidiaries are generally not required separately from the parent/top entity, except where analysis of significant subsidiaries is needed to assess compliance with the Framework, or to highlight limitations on intra-group fund/capital transfers.

Standalone (solo) disclosures are required from individual banks only if they are **not** the top consolidated entity in the group.

Frequency of Disclosures: Banks must make Pillar 3 disclosures **at least half-yearly** (e.g., as of March 31 and September 30), regardless of whether financial statements are audited.

Exceptions (higher frequency required):

- ✓ **Capital Adequacy** disclosures.
- ✓ **Credit Risk: General Disclosures** for all banks.
- ✓ **Credit Risk: Disclosures for Portfolios Subject to the Standardised Approach**. These must be made **at least quarterly** (e.g., March 31, June 30, September 30, December 31).

Required Disclosures: Scope, Capital, and Credit Risk

Table DF-1: Scope of Application

- **Qualitative:** List of group entities considered and not considered for consolidation (accounting vs regulatory scope).
- **Quantitative:** Total balance sheet equity, capital deficiencies in subsidiaries, aggregate amounts of capital deficiencies deducted.

Table DF-2: Capital Adequacy

- **Qualitative:** Summary discussion of approach to assessing capital adequacy.
- **Quantitative:** Capital requirements for credit risk (portfolios/securitisation), market risk (duration approach by interest rate, FX, equity), operational risk (BIA/Standardised), and Tier 1/Total Capital ratios.

Table DF-3: Credit Risk - General Disclosures

- **Qualitative:** Definitions of past due/impaired, credit risk management policy.
- **Quantitative:** Total gross credit risk exposures, geographic distribution, industry type, residual contractual maturity, amount of NPAs (Gross/Net), NPA ratios, movement of NPAs and provisions.

Required Disclosures: Portfolios, Mitigation, and Securitisation

Table DF-4: Credit Risk Portfolios under Standardised Approach

Qualitative:	Quantitative:
• Names of credit rating agencies used, reasons for changes, types of exposure for each agency.	• Exposure amounts after risk mitigation subjected to standardized approach, categorized into three major risk buckets: ◦ Below 100% risk weight, 100% risk weight, ◦ More than 100% risk weight, and ◦ Deducted.

Table DF-5: Credit Risk Mitigation (Standardised Approaches)

Qualitative:	Quantitative:
• Policies and processes for collateral valuation, management of main types of collateral taken, and credit/market risk concentrations.	• Total exposure covered by eligible financial collateral (after haircuts) and covered by guarantees/credit derivatives.

Table DF-6: Securitisation Exposures

Qualitative:	Quantitative:
• Objectives in relation to securitisation, roles played by the bank, processes to monitor changes in credit/market risk, and accounting policies.	• Exposures in banking book vs. trading book, aggregate amount of exposures securitised, retained, or purchased broken down by risk weight bands.

Required Disclosures: Market, Operational, IRRBB, and Counterparty Risks

Table DF-7: Market Risk in Trading Book	
Qualitative: General disclosure requirements for portfolios covered.	**Quantitative:** Capital requirements for interest rate risk, equity position risk, and foreign exchange risk.

Table DF-8: Operational Risk
Qualitative: Disclosure of the approach(es) for operational risk capital assessment for which the bank qualifies.

Table DF-9: Interest Rate Risk in the Banking Book (IRRBB)	
Qualitative: Nature of IRRBB, key assumptions (loan prepayments, non-maturity deposits), and frequency of measurement.	**Quantitative:** The increase (decline) in earnings and economic value for upward and downward rate shocks broken down by currency.

Table DF-10: Counterparty Credit Risk (CCR)	
Qualitative: Methodology used to assign economic capital/limits, policies for securing collateral, and wrong-way risk exposures.	**Quantitative:** Gross positive fair value of contracts, credit derivative transactions to CCR, aggregate amount of exposures.

Q. Which of the following statements are correct in context of the Revised Basel Framework and its implementation in India?

1. The Revised Basel Framework is structured around three Pillars, with Pillar 3 aimed at ensuring capital adequacy by using internal rating models and stress testing.
2. Pillar 2 (Supervisory Review Process) requires banks to consider risks not fully captured or not at all covered under Pillar 1, along with certain external factors.
3. Under Basel norms in India, commercial banks are required to use the Foundation Internal Rating Based (FIRB) Approach for credit risk and Advanced Measurement Approach (AMA) for operational risk.

A) Only statements 1 and 3 are correct
B) Only statements 2 is correct
C) Only statements 1 and 2 are correct
D) All statements (1, 2, and 3) are correct

Answer: B

Solution:

1. Incorrect – Pillar 3 is about market discipline through disclosures, not capital adequacy via internal models.
2. Correct – Pillar 2 ensures risks not captured in Pillar 1, including external factors, are addressed via ICAAP.
3. Incorrect – In India, banks must use the Standardised Approach (SA) for credit risk and Basic Indicator Approach (BIA) for operational risk (not FIRB and AMA).

Q. Which of the following statements regarding the Supervisory Review Process (Pillar 2) under the Basel Framework are correct?

1. The capital adequacy ratio prescribed under Pillar 1 is considered sufficient to cover all types of risks faced by banks.
2. Interest rate risk in the banking book, though part of market risk, may not be fully captured under regulatory CRAR.
3. Strategic risk and reputational risk are both considered equivalent to operational risk under the Supervisory Review Process.

A) Only statements 1 and 3 are correct
B) Only statements 2 and 3 are correct
C) Only statements 1 and 2 are correct
D) All statements (1, 2, and 3) are correct

Answer: B

Explanation:

Statement 1: Incorrect- Pillar 1 covers only minimum regulatory capital for credit, market, and operational risk. It is explicitly stated that additional capital may be needed due to other risks not captured under CRAR.

Statement 2: Correct - Interest rate risk in the banking book is part of market risk, but may not be fully captured under CRAR, hence banks are advised to assess it internally.

Statement 3: Correct- Both strategic risk and reputational risk are mentioned as being equivalent to operational risk in the context of Pillar 2.

Q. Which of the following statements correctly reflect the principles and components of the Supervisory Review Process (SRP) and the Internal Capital Adequacy Assessment Process (ICAAP) as per the Basel Capital Adequacy Framework?

1. The Basel II and III frameworks expanded the scope of capital adequacy by including operational risk under Pillar 1, which was not covered under Basel I.
2. Under Principle 1 of the SRP, banks are required to maintain capital only at the minimum regulatory level, provided their internal risk assessments justify it.
3. Principle 4 of SRP empowers supervisors to mandate immediate corrective actions if a bank's capital falls below the level required to support its risk characteristics.

A) Only statements 1 and 3 are correct
B) Only statements 2 and 3 are correct
C) Only statements 1 and 2 are correct
D) All statements (1, 2, and 3) are correct

Answer: A

Solution:

Statement 1 is correct: Basel II introduced operational risk in capital calculation, which was not part of Basel I.

Statement 2 is incorrect: Principle 1 expects banks to assess overall capital adequacy in relation to their risk profile and maintain a strategy to operate above the minimum, not just at the minimum.

Statement 3 is correct: Principle 4 clearly states that supervisors should intervene early and take rapid remedial action if capital falls below what is required for a bank's risk characteristics.

Q. Which of the following statements correctly reflect key features and responsibilities relating to ICAAP governance and integration?

1. Every banking entity within a group must prepare ICAAP on a solo basis, but consolidation at the group level is optional.
2. The Board of Directors must approve the structure and content of ICAAP to ensure it becomes part of the management culture.
3. Senior management is responsible for submitting ICAAP outcomes to RBI directly, bypassing Board review.

A) Only statements 1 and 3 are correct
B) Only statements 2 and 3 are correct
C) Only statements 2 is correct
D) All statements (1, 2, and 3) are correct

Answer: C

Solution:

Statement 1 is incorrect: ICAAP must be prepared both on a solo basis and at the consolidated level—consolidation is not optional.

Statement 2 is correct: The Board’s approval is crucial to ensure ICAAP is embedded in the bank’s strategic framework.

Statement 3 is incorrect: ICAAP results are submitted to the Board, which then sends them to RBI.

Q. Which of the following statements are true regarding risk identification, internal controls, and MIS within the ICAAP framework?

1. MIS should enable proactive risk management by providing regular, accurate, and timely information on aggregate risk profile.
2. Internal audit teams may assist business lines in designing the ICAAP, provided separation of duties is maintained.
3. The risk management function should be independent of business lines to ensure objectivity and prevent conflict of interest.

A) Only statements 1 and 3 are correct
B) Only statements 2 and 3 are correct
C) Only statements 1 and 2 are correct
D) All statements (1, 2, and 3) are correct

Answer: A

Solution:

Statement 1 is correct: A sound MIS system should deliver real-time and aggregated risk data.

Statement 2 is incorrect: Internal auditors must remain independent and should not assist business lines in ICAAP preparation.

Statement 3 is correct: Independence of risk management is essential to prevent conflict of interest.

Q. In the context of ICAAP being a risk-based process, which of the following statements is not accurate?

1. ICAAP must cover all material risks, including those that are not easily quantifiable like strategic risk.
2. For qualitative risks, the ICAAP should focus on mitigation strategies rather than numerical measures.
3. Banks may omit risks from ICAAP if they are difficult to measure or if data is unavailable.

A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 3 Only

Answer: D

Solution:

Statement 3 is incorrect—no material risk can be omitted simply due to difficulty in measurement. ICAAP requires both quantitative and qualitative assessments depending on the nature of risk.

Q. Which of the following statements is/are true about stress testing and risk aggregation under ICAAP?

1. Stress testing helps evaluate the bank's exposure to extreme yet plausible market conditions.
2. Banks should document assumptions used in risk aggregation and consider the impact of diversification.
3. Scenario analyses are used primarily for daily operational decisions and do not impact ICAAP capital planning.

A. 1 and 2 only

B. 2 and 3 only

C. 1 and 3 only

D. 1, 2, and 3

Answer: A

Solution:

Statements 1 and 2 align with ICAAP expectations—stress tests gauge vulnerabilities under adverse conditions, and aggregation must be systematic with clear documentation.

Statement 3 is wrong—scenario analysis is integral to ICAAP for understanding capital adequacy under stress.

Q. Which of the following statements correctly relate to the structure, purpose, and application of the ICAAP Document as per RBI guidelines?

1. The ICAAP document must include an Executive Summary that compares the bank's internal capital assessment with minimum CRAR requirements under Pillar 1.
2. ICAAP methodology can incorporate capital buffers based on strategic plans or credit rating objectives beyond regulatory minimums.
3. Scenario analyses used in ICAAP must consider events with a frequency of once every 3 years to simulate routine stress.

A) Only statements 1 and 3 are correct
B) Only statements 2 and 3 are correct
C) Only statements 1 and 2 are correct
D) All statements (1, 2, and 3) are correct

Answer: C

Solution:

Statement 1 is correct: The Executive Summary includes a comparison between internal capital and Pillar 1 CRAR.

Statement 2 is correct: ICAAP may include buffers for strategic reasons or credit ratings.

Statement 3 is incorrect: Scenario severity is expected to reflect events occurring once in 25 years.

Q. Which of the following statements are correct regarding the purpose and guiding principles of Pillar 3 disclosures under the Basel framework?

1. The intent of Pillar 3 is to allow banks to attract more capital from private investors through marketing disclosures.
2. Market discipline enhances corporate governance and contributes to banking stability.
3. One of the guiding principles of Pillar 3 is that disclosures should be meaningful to users.

A) Only statements 1 and 3 are correct
B) Only statements 2 and 3 are correct
C) Only statements 1 and 2 are correct
D) All statements (1, 2, and 3) are correct

Answer: B

Solution:

Statement 1 is incorrect – the purpose is not promotional but to enhance transparency for risk assessment.

Statement 2 is correct – market discipline does support corporate governance and banking soundness.

Statement 3 is correct – “meaningful to users” is one of the five guiding principles.

Q. Which of the following statements correctly describe the scope, frequency, and format of Pillar 3 disclosures?

1. Disclosures are required to be made at the individual bank level only when the bank is not the top consolidated entity.
2. Capital Adequacy and certain credit risk disclosures must be made on a quarterly basis.
3. Pillar 3 disclosures can be made only in physical reports submitted to the RBI.

A) Only statements 1 and 3 are correct
B) Only statements 2 and 3 are correct
C) Only statements 1 and 2 are correct
D) All statements (1, 2, and 3) are correct

Answer: C

Solution:

Statement 1 is correct – individual banks that aren’t the top consolidated entity must disclose independently.

Statement 2 is correct – specific items like Capital Adequacy must be disclosed quarterly.

Statement 3 is incorrect – disclosures can be made online or in financial results.

Q. Which of the following are valid principles or requirements under Pillar 3 regarding disclosure governance, validation, and confidentiality?

1. Banks must ensure that all Pillar 3 disclosures are externally audited by default.
2. Proprietary information may be withheld if its disclosure weakens a bank's competitive position.
3. Validation processes should ensure consistency with audited annual financial statements.

A) Only statements 1 and 3 are correct
B) Only statements 2 and 3 are correct
C) Only statements 1 and 2 are correct
D) All statements (1, 2, and 3) are correct

Answer: B

Solution:

Statement 1 is incorrect – external audit is not mandatory unless specifically required.

Statement 2 is correct – proprietary/confidential info can be withheld.

Statement 3 is correct – validation ensures disclosures align with audited figures.

Q. A commercial bank, XYZ Bank Ltd., is assessing its **capital adequacy position under Basel III as of March 31, 2025. The following details are available:**

Tier I Capital: ₹8,000 crore; Tier II Capital: ₹6,000 crore; Subordinated debt included in Tier II: ₹4,500 crore (all above 5 years maturity); Revaluation reserves: ₹1,000 crore; Capital requirement for credit risk: ₹1,200 crore; Capital requirement for market risk: ₹300 crore; Capital requirement for operational risk: ₹400 crore; Investment fluctuation reserve: ₹500 crore; Paid-up equity capital and free reserves form 90% of Tier I; Bank follows Indian regulatory norms (i.e., 9% minimum capital adequacy)

Based on the above information, which of the following statements is/are correct regarding the capital adequacy and composition for XYZ Bank Ltd. under Basel III guidelines?

1. The maximum permissible Tier II capital is equal to Tier I capital, i.e., ₹8,000 crore.
2. Tier III capital can be considered while calculating market risk capital requirement.
3. Total capital available for adequacy computation is ₹14,000 crore 3750 + 5000
4. Risk-weighted assets (RWA) = ₹1,200 crore + 12.5 × ₹300 crore + 12.5 × ₹400 crore = ₹9,950 crore.
5. The Capital to Risk Weighted Assets Ratio (CRAR) of XYZ Bank Ltd. is above the minimum Indian requirement.

140.70

A. 1, 3, 4, and 5 only
B. 1, 2, 3, and 4 only
C. 2, 3, and 4 only
D. 1, 3, and 5 only

Answer: A

Explanation:

Statement 1 is correct: Tier II capital is capped at 100% of Tier I, i.e., ₹8,000 crore maximum.

Statement 2 is incorrect: Tier III capital has been phased out under Basel III; it was applicable under Basel II for market risk.

Statement 3 is correct: Total capital = Tier I (₹8,000 crore) + Tier II (₹6,000 crore) = ₹14,000 crore.

Statement 4 is correct: RWA = ₹1,200 + (12.5×300) + (12.5×400) = ₹1,200 + ₹3,750 + ₹5,000 = ₹9,950 crore.

Statement 5 is correct: CRAR = (₹14,000 / ₹9,950) × 100 ≈ 140.7%, which is well above the 9% requirement in India.

Q. Which of the following statements regarding **the Supervisory Review Process (Pillar 2) under the Basel Framework are correct?**

1. The capital adequacy ratio prescribed under Pillar 1 is considered sufficient to cover all types of risks faced by banks.
2. Interest rate risk in the banking book, though part of market risk, may not be fully captured under regulatory CRAR.
3. Strategic risk and reputational risk are both considered equivalent to operational risk under the Supervisory Review Process.
4. Under the ICAAP, banks must use a uniform market consensus approach to assess capital adequacy and future requirements.
5. Risks such as climate risk, settlement risk, and model risk may require banks to maintain additional capital beyond the Pillar 1 minimum.

A. Only 2 and 5 are correct
B. Only 2, 3, and 5 are correct
C. Only 1, 3, and 4 are correct
D. Only 2, 3, 4, and 5 are correct

Answer: B

Explanation:

Statement 1: Incorrect- Pillar 1 covers only minimum regulatory capital for credit, market, and operational risk. It is explicitly stated that additional capital may be needed due to other risks not captured under CRAR.

Statement 2: Correct - Interest rate risk in the banking book is part of market risk, but may not be fully captured under CRAR, hence banks are advised to assess it internally.

Statement 3: Correct- Both strategic risk and reputational risk are mentioned as being equivalent to operational risk in the context of Pillar 2.

Statement 4: Incorrect- There is no single or uniform market consensus on how to conduct ICAAP. It is explicitly noted that different approaches are acceptable.

Statement 5: Correct- Risks like climate risk, payment & settlement risk, and model risk are part of risks that may require additional capital under ICAAP.

Q. Which of the following statements correctly reflect **key features and responsibilities relating to ICAAP governance and integration?**

1. Every banking entity within a group must prepare ICAAP on a solo basis, but consolidation at the group level is optional.
2. The Board of Directors must approve the structure and content of ICAAP to ensure it becomes part of the management culture.
3. Senior management is responsible for submitting ICAAP outcomes to RBI directly, bypassing Board review.
4. The strategic plan forming part of ICAAP must include current and future capital needs, capital expenditure, and desired capital levels.
5. The integration of ICAAP can include its use in pricing decisions, credit approval, and internal capital allocation.

A. 1, 2, 3
B. 2, 4, 5
C. 3, 4, 5
D. 1, 4, 5

Answer: B

Solution:

Statement 1 is incorrect: ICAAP must be prepared both on a solo basis and at the consolidated level—consolidation is not optional.

Statement 2 is correct: The Board’s approval is crucial to ensure ICAAP is embedded in the bank’s strategic framework.

Statement 3 is incorrect: ICAAP results are submitted to the Board, which then sends them to RBI.

Statement 4 is correct: The strategic plan under ICAAP must include capital planning elements.

Statement 5 is correct: ICAAP integration can influence credit decisions, pricing, and internal capital allocation.

Q. Identify the correct statements related to **capital adequacy and risk-related disclosures banks are required to make.**

1. The bank must disclose capital requirements separately for credit, market, and operational risks.
2. Banks must provide a qualitative summary of their capital planning approach to support current and future activities.
3. Credit risk disclosures must include both fund-based and non-fund-based exposures, classified by geography and residual maturity.
4. The capital adequacy ratios (CET 1, Tier 1, and Total) must be disclosed only at the standalone entity level.
5. Movement of provisions and NPAs must be shown in detail, including write-offs and recoveries impacting the income statement.

A. 1, 3, and 4 only
B. 2, 3, and 5 only
C. 1, 2, 3, and 5 only
D. 2, 3, 4 and 5 only

Answer: C

Solution:

1 is correct – Basel III disclosures segment capital requirements by risk category.

2 is correct – A qualitative capital adequacy strategy must be disclosed.

3 is correct – Detailed breakdown by geography, maturity, and exposure type is essential.

4 is incorrect – Capital adequacy ratios must be disclosed both at consolidated group level and for significant subsidiaries.

5 is correct – Movement of provisions and direct income statement impacts must be separately disclosed.

UNIT 28

Asset Classification and Provisioning Norms

STRUCTURE

Defining the Non-Performing Asset (NPA)

Core Definition: An asset ceases to generate income for the bank. Banks should classify an account as NPA only if the interest charged during any quarter is not serviced fully within 90 days from the end of the quarter.

Asset Type	NPA Classification Trigger
Term Loans	**Interest** and/or principal installment remains overdue for **more than 90 days**.
Overdraft/Cash Credit (OD/CC)	The account remains **'out of order'**.
Bills Purchased/Discounted	The bill remains overdue for more than **90 days**.
Agricultural Advances	Installment/interest overdue for two crop seasons (short duration crops) or one crop season (long duration crops).
Securitisation	Liquidity facility outstanding for **> 90 days**.
Derivative Transactions	**Overdue receivables** representing positive mark-to-market (MTM) value remain unpaid for 90 days from the specified due date.

The Asset Classification Framework

CRITICAL RULE: Classification must be Borrower-wise and not Facility-wise. If one facility of a borrower becomes an NPA, all other funded facilities granted to that client must also be classified as non-performing.

Standard Assets

Does not disclose any problems and does not carry more than normal risk attached to the business.

Substandard Assets

Has remained an NPA for a period less than or equal to 12 months.

Doubtful Assets

Has remained in the substandard category for a period of 12 months.

Loss Assets

Loss has been identified by the bank or internal/external auditors, or the RBI inspection, but the amount has not been written off wholly.

Provisioning Norms: Substandard and Loss Assets

15% / 25%

Substandard Assets:

- A general provision of 15% on total outstanding balance is required.
- Unsecured Exposures: Attract an additional provision of 10% (total 25%). 'Unsecured exposure' is where realisable value of security is not more than 10% ab-initio of the outstanding exposure.
- Infrastructure loans with escrow accounts attract 20% instead of 25%.

100%

Loss Assets:

- Should be completely written off.
- If permitted to remain on the books for any reason, 100% of the outstanding balance should be provided for.

Provisioning Norms: Doubtful Assets

Unsecured Portion: 100% provision is required immediately to the extent the advance is not covered by the realisable value of the security.

Period for which advance remained 'doubtful'	Standard Provision Requirement (Secured)	Accelerated Penalty Provision (If SMA reporting fails)
Up to one year	25%	40%
One to three years	40%	100%
More than three years	100%	100%

Accelerated Provisioning Penalties: If banks fail to report SMA to CRILC, accelerated provisions apply. Example: Sub-standard unsecured jumps from 25% to 40% (6 months to 1 year). Doubtful 1st year secured jumps from 25% to 40%.

NotebookLM

Provisioning Norms: Standard Assets

Provisions on standard assets are not reckoned for arriving at net NPAs and must be shown separately as "Contingent Provisions against Standard Assets".

0.25%
Agriculture and SME sectors

1.00%
Commercial Real Estate (CRE) Sector

0.75%
CRE - Residential Housing (CRE - RH)

2.00%
Housing loans extended at teaser rates (reverts to 0.40% after 1 year of higher rates)

5%
Restructured advances

0.40%
All other loans and advances (Including Medium Enterprises)

Standard Assets

(i) Banks are required to make general provision for standard assets at the following rates for the funded outstanding on global loan portfolio basis:

(a) Farm Credit to agricultural activities and Small and Micro Enterprises (SMEs) sectors at 0.25 per cent;

(b) advances to Commercial Real Estate (CRE)1 Sector at 1.00 per cent;

(c) advances to Commercial Real Estate – Residential Housing Sector (CRE - RH)2 at 0.75 per cent

(d) housing loans extended at teaser rates at 2.00 per cent the provisioning on these assets would revert to 0.40 per cent after 1 year from the date on which the rates are reset at higher rates if the accounts remain 'standard'.

(e) restructured advances – as stipulated in the prudential norms for restructuring of advances.

(g) Advances restructured and classified as standard in terms of the Master Direction – Reserve Bank of India (Relief Measures by Banks in Areas affected by Natural Calamities) Directions 2018 – SCBs, as updated from time to time, at 5%.

(h) All other loans and advances not included in (a) – (f) above at 0.40 per cent.

Exceptions and Temporary Deficiencies

Asset classification must be based on the record of recovery, not merely temporary deficiencies like non-availability of adequate drawing power or temporary limit excesses.

Stock Statements

Drawing power is calculated from current stock statements. Outstanding balances based on stock statements older than three months are deemed irregular.

Working Capital Default

A working capital borrowal account becomes an NPA if irregular drawings are permitted for a continuous period of 90 days, even if the borrower's financial position is otherwise satisfactory.

Unrenewed Limits

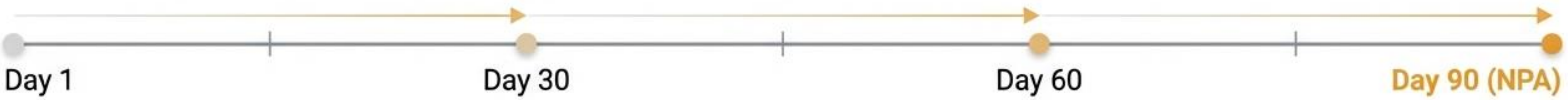
Regular and ad hoc credit limits must be reviewed/regularised within three months. An account where limits limits are not reviewed/renewed within 180 days from the due date is treated as an NPA.

Early Warning Signals: SMA and CRILC

CRILC Reporting Mandate

The RBI mandates banks to report credit information, including SMA classification, to the **Central Repository of Information on Large Credits** (CRILC) on all borrowers with an aggregate exposure of ₹50 million and above. Main report is monthly; default report is weekly (every Friday).

SMA Sub-categories (Term Loans)	SMA Sub-categories (Revolving Credit - OD/CC)
SMA-0: Principal/interest not overdue for more than 30 days, but account shows signs of incipient stress. **SMA-1**: Overdue between 31 – 60 days. **SMA-2**: Overdue between 61 – 90 days.	**SMA-1**: Outstanding balance continuously exceeds limit/drawing power for 31-60 days. **SMA-2**: Exceeds limit continuously for 61-90 days.



Income Recognition & Accelerated Penalties

Income Recognition Shift

- Income from NPAs cannot be recognised on an accrual basis; it must be strictly on a cash basis (only upon realization).
- Reversals: Interest wrongly recognized in the past must be reversed. If recognized in previous years, a provision for an equivalent amount must be made. Funded interest is recognized only when cash is received.

Resolution Plan (RP) Timeline Penalties

180 days from end of Review Period

20% additional provision as % of total outstanding.

365 days from commencement of Review Period

15% additional (Total 35% additional provisioning).

If a viable RP is not implemented within strict timelines, additional provisions are mandated.

The Path to Recovery: Upgradation Rules

If arrears of interest and principal are paid by the borrower, the account should no longer be treated as non-performing and may be classified as 'standard'.

MSME Accounts (Exposure **< ₹25 crores**)	All Other Accounts
Must demonstrate 'Satisfactory Performance' during a 'Specified Period' (one year from the commencement of the first payment of interest/principal on the facility with the longest moratorium).	Upgraded only when all outstanding loan facilities demonstrate satisfactory performance up to the date by which **at least 10% repayment** of the sum of outstanding principal debt (per the RP) and capitalized interest is repaid.

Sector-Specific NPA Guidelines

Agricultural Advances: Tied to crop seasons determined by the State Level Bankers' Committee (SLBC). Short duration crops = overdue 2 seasons. Long duration crops = overdue 1 season.

Government Guaranteed Advances: Credit facilities backed by Central Government guarantees are treated as NPA for income recognition but are exempt from asset classification/provisioning unless the guarantee is repudiated by the Government. State Government guarantees attract normal normal norms if overdue > 90 days.

Project Loans: Divided into infrastructure and non-infrastructure. Exemptions from standard NPA classification apply if project completion is delayed due to legal/government approvals beyond promoters' control, prior to commencement of commercial operations.

Resolution Mechanism: Insolvency and Bankruptcy Code (IBC)

Core Shift: The IBC transitioned the Indian insolvency regime from **‘debtor-in-possession’** to **‘creditor-in-control’**.

Triggered by Financial/Operational Creditors or Debtors → Moratorium Enacted (Parallel proceedings stopped) → Managed by certified Insolvency Professionals → Strict deadline of 180 days (extendable to 330 days maximum)

Liquidation Waterfall Priority

Workers’ salaries (up to 24 months) receive first priority in case of liquidation of assets of a company, ahead of secured creditors. This priority ensures payment before other debts, including secured creditors, during the distribution of liquidation proceeds.

Systemic Resolution: The Concept of 'Bad Banks'

Purpose: To acquire non-performing assets from commercial banks, easing balance sheet burden and freeing them to lend again, not necessarily to generate profit.

Backed by a Government guarantee of ₹30,600 crore to cover shortfalls if the asset sells for less than expected.

Q. Which of the following statements are correct with respect to the classification of Non-Performing Assets (NPAs)?

1. A substandard asset is one which has remained non-performing for more than 12 months and shows well-defined credit weaknesses that may lead to partial loss.
2. A doubtful asset is one which has remained in the substandard category for a continuous period of at least 12 months and its recovery is highly improbable under currently known conditions.
3. A loss asset is considered uncollectible, and its value is so impaired that its continuance as a bankable asset is unwarranted, even though some recovery may still be possible.

A. Only 2 and 3
B. Only 1 and 2
C. Only 1 and 3
D. All 1, 2 and 3

Answer: A

Explanation:

Statement 1 is incorrect: A substandard asset is one that has remained NPA for less than or equal to 12 months, not more than 12 months.

Statement 2 is correct: Doubtful assets are those that have remained substandard for 12 months and exhibit weaknesses that make recovery highly uncertain.

Statement 3 is correct: A loss asset is one identified as uncollectible and of negligible value for continuing as a bankable asset, though minor recovery may be possible.

Q. Which of the following statements correctly align with the RBI's asset classification norms related to borrower-wise NPA recognition and treatment of derivative exposures?

1. If a single loan facility extended to a borrower becomes irregular, only that facility shall be classified as NPA, without affecting other loans or investments made to the same borrower.
2. When a derivative contract shows a positive mark-to-market (MTM) value and remains unpaid for over 90 days, it becomes an NPA, and all other funded exposures of the borrower are also classified as NPAs—except where such derivatives were entered into between April 2007 and June 2008.
3. If documents under a Letter of Credit (LC) are dishonoured and the borrower does not reimburse the bank, the discounted bills under such LC shall be classified as NPA from the date other credit facilities of the borrower were classified as NPA.

A) Only statements 1 and 3 are correct
B) Only statements 2 and 3 are correct
C) Only statements 1 and 2 are correct
D) All statements (1, 2, and 3) are correct

Answer: B

Solution:

Statement 1 – Incorrect: The RBI mandates borrower-wise classification. If one facility is NPA, all exposures to that borrower become NPAs.

Statement 2 – Correct: Positive MTM overdue for 90 days becomes NPA and affects other facilities, except for derivatives entered into between April 2007–June 2008, which are treated separately.

Statement 3 – Correct: If payment under LC fails and the borrower does not pay, such discounted bills are treated as NPAs from the same date as other NPAs.

Q. Which of the following statements relating to the Insolvency and Bankruptcy Code, 2016 (IBC) are correct?

1. The IBC shifted India's insolvency framework from a debtor-in-possession model to a creditor-in-control model, allowing creditors to manage the distressed entity.
2. Under the IBC, only financial creditors can initiate the Corporate Insolvency Resolution Process (CIRP), excluding operational creditors and corporate debtors.
3. A moratorium under the IBC prevents legal action against the corporate debtor during CIRP but does not shield its key managerial personnel.

A) Only statements 1 and 3 are correct
B) Only statements 2 and 3 are correct
C) Only statements 1 and 2 are correct
D) All statements (1, 2, and 3) are correct

Answer: A

Solution:

Statement 1 – Correct: IBC introduced the creditor-in-control model replacing the earlier debtor-in-possession approach, allowing creditors to appoint professionals to manage the debtor's operations during CIRP.

Statement 2 – Incorrect: The CIRP can be initiated not just by financial creditors but also by operational creditors and corporate debtors themselves.

Statement 3 – Correct: A moratorium is imposed upon admission of the insolvency petition which halts legal action against the debtor but does not extend protection to key managerial personnel responsible for insolvency.

Q. Consider the following statements regarding **provisioning norms for advances. Identify which of them are correct:**

1. Loss assets must be written off or provided for to the extent of 100% if retained in the books.
2. Doubtful assets require 100% provisioning for the portion not covered by realisable security value.
3. For doubtful assets secured for more than 3 years, the provisioning required is ~~25%~~. 100%
4. Substandard unsecured assets require a total of 25% provisioning.
5. Infrastructure loans classified as substandard require 20% provisioning if proper escrow mechanism exists. ≤ 10%
6. Unsecured exposures where the realisable security value is ~~15%~~ of the exposure are treated as secured.
7. Standard assets in Commercial Real Estate – Residential Housing (CRE-RH) attract 0.75% provisioning.
8. Advances restructured under RBI relief measures for natural calamities are provisioned at 5%.
9. Housing loans at teaser rates are provisioned at 2% initially, and reduced to 0.40% after 1 year from reset if they remain standard.

Which of the above statements are correct?

A. 1, 2, 4, 5, 7, 8, 9

B. 1, 2, 3, 4, 5, 6

C. 2, 4, 5, 6, 7, 8, 9

D. 1, 2, 4, 5, 6, 7, 9

Answer: A

Solution:

- ✓ Statement 1: Correct — Loss assets must be written off or 100% provisioned.
- ✓ Statement 2: Correct — For doubtful assets, 100% provisioning is required for unsecured parts.
- ✓ **Statement 3: Incorrect** — Provisioning is 100% for secured portion if doubtful for more than 3 years.
- ✓ Statement 4: Correct — Substandard unsecured exposures require total of 25% provisioning.
- ✓ Statement 5: Correct — Infrastructure loans with escrow mechanism require only 20% provisioning.
- ✓ **Statement 6: Incorrect** — If realisable value is ≤10%, the exposure is unsecured. At 15%, it is considered secured.
- ✓ Statement 7: Correct — CRE-RH loans require 0.75% provisioning.
- ✓ Statement 8: Correct — Restructured advances under natural calamity relief attract 5% provisioning.
- ✓ Statement 9: Correct — Teaser rate loans attract 2% initially, reduced to 0.40% if standard after reset.

Q. A bank has the following credit exposures (all figures in ₹ lakh). Compute the total provisioning requirement as per RBI norms:

Account A – Substandard (Unsecured exposure)

Outstanding: ₹240 lakh @ 25% → 60L

Account B – Doubtful > 3 years

Outstanding: ₹300 lakh → 300L

Realisable security value (secured portion): ₹120 lakh

Account C – Doubtful for 2 years (i.e., 1–3 years)

Outstanding: ₹180 lakh

Realisable security value: ₹100 lakh 120L

180
− 100 @ 40%
80

Account D – Standard Asset – CRE Exposure

Outstanding: ₹5000 lakh → 50L

Account E – Standard Asset – Unhedged Forex Exposure

Outstanding: ₹4000 lakh

Likely Loss/EBID = 40% .80 x 4K = 32L .4 + .4

562L

What is the total provision that the bank must make?

A. ₹ 532 lakh

B. ₹ 562 lakh

C. ₹ 632 lakh

D. ₹ 662 lakh

Answer: B

Solution:

1. Account A – Substandard (Unsecured)

Unsecured substandard provisioning = 25% of outstanding (only 10% security → unsecured); Provision = 25% × 240 = ₹60 lakh

2. Account B – Doubtful > 3 years

Unsecured portion = 300 – 120 = 180 lakh → 100% provision = 180

Secured portion = 120 lakh → provision = 100% (because >3 yrs)

Provision = 180 + 120 = ₹300 lakh

3. Account C – Doubtful for 2 years (1–3 years bucket)

Outstanding = 180; Security = 100; Unsecured portion = 80 → 100% = 80; Secured portion = 100 → 40% 3 yrs) = 40 → @40%.

Total = ₹120 lakh

120L

4. Account D – Standard – CRE Exposure

Provision = 1% of 5000 = ₹50 lakh

5. Account E – Standard – Unhedged Forex Exposure

Base standard provision = 0.40%

Incremental provision for Likely Loss/EBID = 40 bps (0.40%) because level = 45%

Total provision rate = 0.40% + 0.40% = 0.80%

Provision = 0.80% × 4000 = ₹32 lakh

Total Provision Required = 60 + 300 + 120 + 50 + 32 = ₹562 lakh

Q. Below are the statements related to **Provisioning for Country Risk. Select the correct set of statements.**

1. Banks must make provisions on net funded country exposures on a graded scale ranging from 0.25% to 100%.
2. The provisioning requirement for a country under the "Moderate" (B1) category is 5%.
3. For countries classified as "Restricted" (C2), the provisioning requirement is 25%.
4. If a country's net funded exposure is 1% or more of a bank's total assets, it must make provisions for country risk.
5. Banks can use their provisioning buffer for specific provisions for NPAs during a system-wide downturn with the prior approval of the RBI.
6. Banks are required to disclose their Provisioning for Country Risk (PCR) in the Cash Flow Statement.
7. The provisioning for a country under the "Low" (A2) classification is 0.5%.

A) Statements 1, 2, 4, 5, 7 are correct

B) Statements 2, 3, 5, 6 are correct

C) Statements 1, 4, 5, 7 are correct

D) Statements 1, 2, 4, 5 are correct

Answer: D

Solution:

- ✓ Statement 1 is correct: Banks must make provisions on a graded scale from 0.25% to 100%.
- ✓ Statement 2 is correct: The provisioning requirement for a "Moderate" (B1) category is 5%.
- ✓ Statement 4 is correct: Banks must make provisions if the net funded exposure is 1% or more of total assets.
- ✓ Statement 5 is correct: The provisioning buffer can be used for NPAs during a system-wide downturn, with RBI's prior approval.
- ✓ **Statement 3 is incorrect:** "Restricted" (C2) classification requires 100% provisioning, not 25%.
- ✓ **Statement 6 is incorrect:** PCR must be disclosed in the Notes to Accounts, not the Cash Flow Statement.
- ✓ **Statement 7 is incorrect:** The "Low" (A2) classification has a 0.25% provisioning, not 0.5%.

Q. A statutory auditor is reviewing the asset classification of various borrower accounts in XYZ Bank Ltd. for the financial year ending March 31, 2025. The auditor encounters the following borrower accounts and seeks to determine whether they are to be classified as Non-Performing Assets (NPA) as per RBI norms:

1. Account D: A bill purchased and discounted on December 15, 2024, remains unpaid until January 25, 2025.
2. Account E: A receivable from a derivative transaction with a positive mark-to-market value was due on December 15, 2024, but payment has not been received until March 25, 2025.

Which of the above accounts should be classified as NPAs as of March 31, 2025, according to RBI norms?

A. Only Statement 1 is correct

B. Only Statement 2 is correct

C. Both statements are correct

D. Neither 1 nor 2 is correct

Correct Answer: B. Only Statement 2 is correct
Explanation:
Let's analyze both accounts as per **RBI's NPA classification norms**
Account D – Bill Purchased & Discounted
Date of bill: **15 Dec 2024**
Still unpaid: **25 Jan 2025**
For bills purchased/discounted, **overdue > 90 days** from the **due date** is required to classify as NPA.
If the bill is unpaid only till **25 Jan**, that's about **40 days overdue**, so **it is NOT yet NPA** as of **31 Mar 2025** (unless it crosses 90 days).
Conclusion: Account D → **Not NPA**

Account E – Derivative Receivable
Due date: **15 Dec 2024**
Still unpaid: **25 Mar 2025**
For **derivative receivables**, RBI mandates classification as **NPA if overdue > 90 days**.
From 15 Dec 2024 to 25 Mar 2025 = **> 90 days overdue**, so it **must be classified as NPA** by 31 Mar 2025.

Q.A bank is conducting an asset classification audit for several loan accounts under different scenarios. Evaluate the treatment and classification of the following loan accounts and identify the correct set of observations:

- Account A: A borrower under a consortium arrangement remits all repayments to the lead bank, which fails to distribute funds to other member banks. Bank Y, one of the member banks, has not received its share for the last 90 days.
- Account B: A commercial bank sanctioned an advance to a PACS for short-duration crops. One of the facilities has been in default for more than two crop seasons. Other facilities remain current.
- Account C: A borrower has committed fraud, and the realisable value of the security has dropped to 8% of the outstanding dues. The bank is evaluating provisioning treatment and classification.
- Account D: A term loan is backed by an NSC, with a sufficient margin maintained. The borrower has not made repayments in the last 150 days.
- Account E: A housing loan was extended to a staff member with a special arrangement: interest recovery begins after the full principal has been repaid. Interest has been debited quarterly for the last year but not paid.

Which of the following sets of classifications and treatments is correct?

A. Account A- NPA; Account B – classify only the defaulted facility as NPA; Account C – Loss Asset; Account D – not NPA; Account E – not NPA.
B. Account A – Standard; Account B – All facilities NPA; Account C – Doubtful Asset; Account D – NPA; Account E – NPA.
C. Account A – NPA; Account B – Entire exposure classified as NPA; Account C – Doubtful Asset; Account D – NPA; Account E – NPA.
D. Account A – Standard; Account B – Only defaulted facility as NPA; Account C – Loss Asset; Account D – NPA; Account E – NPA.

Answer: A

Solution:

Account A: As per consortium guidelines, if Bank Y has not received its share and no express consent exists from the lead bank, the account is treated as not serviced in Bank Y's books and hence classified as NPA.

Account B: Only the particular facility in default after two crop seasons is treated as NPA, not the entire exposure.

Account C: If realisable value of security is <10%, the security is ignored, and the account is straightaway classified as a Loss Asset.

Account D: Advances against NSCs with adequate margin are not treated as NPA, regardless of repayment delay.

Account E: For staff housing loans with interest recovery deferred till principal repayment, interest isn't overdue until due date—not classified as NPA until default on due dates.

Q. A bank has identified the following irregularities in a large consortium advance account as on 28th May 2026:

- Total outstanding exposure: ₹48.80 crore
- ECGC cover available: 12% of total exposure → 5.86 Cr
- Realisable value of eligible security (after haircut prescribed by RBI-approved valuer): ₹11.45 crore
- Estimated legal recovery expenses: ₹0.95 crore
- Interest suspense not reversed till date: ₹1.75 crore

During RBI inspection, the account was classified as a Loss Asset. @100%.

What amount of provision the bank should create in the current year?

(A) ₹31.45 crore

(B) ₹34.194 crore

(C) ₹36.35 crore

(D) ₹39.10 crore

10.50
5.86
+ 16.37

11.45
− .95
10.50

Rel.
11.45
34.18

48.80
−16.37
32.43
+ 1.75

Answer: B

Solution:

For a Loss Asset, provisioning requirement = 100% of the amount remaining unrecovered and retained in books.

Step 1: Compute effective realisable security value

Eligible security value = ₹11.45 crore; Less: Legal recovery expenses = ₹0.95 crore

Net realisable security = ₹10.50 crore

Step 2: Compute ECGC recoverable amount

ECGC cover = 12% × ₹48.80 crore = ₹5.856 crore

Step 3: Total expected recoveries

= ₹10.50 crore + ₹5.856 crore = ₹16.356 crore

Step 4: Gross unsecured loss portion

= ₹48.80 crore − ₹16.356 crore = ₹32.444 crore

Step 5: Add unreversed interest suspense

Since unrealised interest still remains unprovided: = ₹32.444 crore + ₹1.75 crore = ₹34.194 crore

UNIT 29 Liquidity Management

STRUCTURE

Introduction: The Dual Objectives of ALM

The Everyday Reality

Unlike other risks (interest rate, market, operational), liquidity risk is a normal aspect of everyday management.

The Extreme Threat

Only in extreme cases do liquidity problems translate into fundamental solvency problems.

The Regulatory Response

Following global market turmoil, Basel III introduced specific ratios: the Liquidity Coverage Ratio (LCR) and the Net Stable Funding Ratio (NSFR).

Core Concept: Defining Bank Liquidity

Liquidity represents the ability to accommodate the decreases in liability and to fund the increases in assets. A bank has adequate liquidity when it can obtain sufficient funds either by increasing liabilities or by converting assets, promptly and at a reasonable cost.

Determinants of Needs

Variability of loan demand and Variability of deposits.

The Price of Liquidity

It is a function of market conditions and market perceptions of risks, including both interest rate and credit risks.

Consequences of Failure

If liquidity needs are not met through liquid asset holdings, a bank may be forced to restructure or acquire additional liabilities under adverse market conditions.

Classification: Types of Liquidity Risks

Funding Risk

Definition: Need to replace net outflows due to unanticipated withdrawal (pre-mature closure of deposits) or non-renewal of deposits.

Causes: Fraud causing substantial loss, systemic risk, loss of confidence, liabilities in foreign currencies.

Time Risk

Definition: Need to compensate for non-receipt of expected inflows of funds.

Causes: Severe deterioration in asset quality, standard assets turning into NPAs / borrower defaults, temporary problems in recovery, time involved in managing liquidity.

Call Risk

Definition: Crystallisation of contingent liabilities and inability to undertake profitable business initiatives when desirable.

Causes: Conversion of non-fund based limits into fund-based limits, swaps, and options.

Dimensions and Role of Liquidity Risk Management

Purpose of Effective Management

- Demonstrates to the marketplace that the bank is safe and capable of repaying borrowings.
- Enables the bank to meet prior loan commitments.
- Avoids the unprofitable sale of assets at fire-sale prices.
- Lowers the size of the default risk premium the bank must pay for funds.

Factors Affecting Liquidity Events

- Decline in earnings
- Increase in non-performing assets (NPAs)
- Deposit concentrations
- Downgrading by rating agencies
- Expanded business opportunities or Acquisitions

Methods to Satisfy Funding Needs

- Dispose of liquid assets
- Increase short-term borrowings
- Decrease holdings of less liquid assets
- Increase liabilities of a term nature
- Increase capital funds

Measuring and Managing Liquidity Risk

Measuring and managing liquisity are among the most vital activities of commercial banks. It can be done through two primary approaches.

0|7|1|2|4|

Stock Approach

- **Concept:** Based on the level of assets and liabilities, as well as off-balance sheet exposures, on a particular date.
- **Method:** Evaluates the liquidity position using a series of calculated financial ratios.

Flow Approach

- **Concept:** The basic approach being followed by Indian banks, focusing on measuring and managing net funding requirements.
- **Method:** Involves the preparation of a structural liquidity gap report based on the residual maturities of assets and liabiliities across various time buckets.

The Flow Approach: Net Funding Requirements

Core Mechanism: The Structural Liquidity Gap Report

Calculated based on the residual maturities of assets and liabilities. These residual maturities represent the net cash flow (difference between outflow and inflow) in future time buckets.

Time Bucket 1 → Time Bucket 2 → Time Bucket 3

The Gap

Cumulative gap is calculated at various time buckets. A net funding requirement (shortfall) at a particular time will be represented by a gap that the bank must manage.

Analytical Tools: This analysis requires the construction of a Maturity Ladder and the calculation of a cumulative net excess or deficit under Alternative Scenarios.

Analytical Tools: The Maturity Ladder

DAY-1 Cash Inflows (Total: 240)

Maturing assets: 100
Interest receivable: 80
Asset sales: 50
Drawdown on com. lines: 10

DAY-1 Cash Outflows (Total: 100)

Maturity liabilities: 50
Interest payable: 10
Other deposit run offs: 30
Drawdown on com. lines: 10

Liquidity Excess: 140

The ladder allows a bank to allocate cash inflow/outflow to a given calendar date, establishing a starting point to measure future excess or shortfall.

Stress Testing: Alternative Scenarios

Normal Business Conditions	Institution Specific Crisis	General Market Crisis
Total Inflows: 240 Total Outflows: 100 Result: **Excess 140**	**Key Changes**: Drawdowns drop to 0. Deposit run-offs spike to 140. Total Inflows: 240 Total Outflows: 250 Result: **Shortfall (10)**	**Key Changes**: Asset sales drop to 0. Interest receivable drops. Total Inflows: 145 Total Outflows: 170 Result: **Shortfall (25)**

Exam Note Callout: Notice how asset liquidity vanishes in a general market crisis, while deposit flight defines an institution-specific crisis.

Complexities: Treatment of Foreign Currencies

The Challenge:

In market concerns, foreign liability holders may not distinguish rumors from facts, and domestic liquidity cannot always easily meet foreign funding requirements.

1. Completely Centralise:

Head office manages liquidity for the whole bank in every currency.

2. Decentralise:

Operating divisions are assigned responsibility for their own liquidity (subject to head office limits).

3. Hybrid (Home Currency Responsibility):

A bank assigns responsibility for liquidity in the home currency and overall coordination to the home office, but foreign branch manages the specific foreign currency it issues.

Governance: Setting Tolerance Levels and Limits

Bank management (or supervisors) must set and review quantitative limits to ensure liquidity.

- Cumulative cash flow mismatches (net funding requirement as a percentage of total liabilities over specific periods like next day, next week, etc.).
- Liquid assets as a percentage of short-term liabilities (assets must have a ready market even in stress periods).
- Limit on loan to deposit ratio.
- Limit on loan to capital ratio.

Practical Example: Managing mismatch levels so that residual maturity mismatches for time buckets of 1-14 days and 15-28 days remain around 80% of cash outflows.

Setting Tolerance Level and Limit for Liquidity Risk

Bank's management should set limits to ensure liquidity and these limits should be reviewed by supervisors. Alternatively, supervisors may set the limits. Limits could be set on the following:

1. The cumulative cash flow mismatches (i.e., the cumulative net funding requirement as a percentage of total liabilities) over particular periods – next day, next week, next fortnight, next month, next year. These mismatches should be calculated by taking a conservative view of marketability of liquid assets, with a discount to cover price volatility and any drop in price in the event of a forced sale, and should include likely outflows as a result of draw-down of commitments, etc.
2. Liquid assets as a percentage of short-term liabilities. The assets included in this category should be those which are highly liquid, i.e., only those assets which are judged to be having a ready market even in periods of stress.
3. A limit on loan to deposit ratio.
4. A limit on loan to capital ratio.
5. A general limit on the relationship between anticipated funding needs and available sources for meeting those needs.
6. Primary sources for meeting funding needs should be quantified.
7. Flexible limits on the percentage reliance on a particular liability category, (e.g., certificates of deposits or high cost deposits should not account for more than a certain percentage of total liabilities).
8. Limits on the dependence on individual customers or market segments for funds in liquidity position calculations.
9. Flexible limits on the minimum/maximum average maturity of different categories of liabilities.
10. Minimum liquidity provision to be maintained to sustain operations.

Stock Approach (to Measuring and Managing Liquidity)

Stock approach is based on the level of assets and liabilities as well as off-balance sheet exposures on a particular date. The following ratios are calculated to assess the liquidity position of a bank.

(a) *Ratio of Core Deposit to Total Assets – Core Deposit/Total Assets:* The higher the ratio, the better it is because core deposits are treated to be a stable source of liquidity. Core deposit will constitute deposits from the public received in the normal course of business.

(b) *Net Loans to Totals Deposits Ratio – Net Loans/Total Deposits:* It reflects the ratio of loans to public deposits or core deposits. Total loans in this ratio represent net advances after deduction of provision for loan losses and interest suspense account. Loan is treated to be a less liquid asset and therefore, the lower the ratio, the better it is.

(c) *Ratio of Time Deposits to Total Deposits – Time Deposits/Total Deposits:* Time deposits provide a stable level of liquidity and negligible volatility. Therefore, the higher the ratio, the better it is.

(d) *Ratio of Volatile Liabilities to Total Assets – Volatile Liabilities/Total Assets:* Volatile liabilities like market borrowings are to be assessed and compared with the total assets. Higher portion of volatile assets will cause higher problems of liquidity. Therefore, the lower the ratio, the better it is.

(e) *Ratio of Short-Term Liabilities to Liquid Assets – Short-term Liabilities/Liquid Assets*: Short-term liabilities are required to be redeemed at the earliest. Therefore, they will require ready liquid assets to meet the liability. It is desirable to be lower in the interest of liquidity.

The Stock Approach: Key Liquidity Ratios

1	Core Deposits to Total Assets Ratio	Higher ↑ = Better (stable source of liquidity)
2	Net Loans to Total Deposits Ratio	Lower ↓ = Better (reflects ratio of loans to public deposits)
3	Ratio of Time Deposits to Total Deposits	Higher ↑ = Better (stable level of liquidity)
4	Ratio of Volatile Liabilities to Total Assets	Lower ↓ = Better (high portion causes liquidity problems)
5	Ratio of Short-Term Liabilities to Liquid Assets	Lower ↓ = Better (indicates less immediate pressure)
6	Ratio of Liquid Assets to Total Assets	Higher ↑ = Better (ensures comfortable liquidity)
7	Ratio of Prime Asset to Total Asset	Higher ↑ = Better (includes cash balances with banks)
8	Ratio of Market Liabilities to Total Assets	Lower ↓ = Better

Flow Approach (to Measuring and Managing Liquidity)

The framework for assessing and managing bank liquidity through flow approach has three major dimensions:

(a) Measuring and managing net funding requirements
(b) Managing market access
(c) Contingency planning

Liquidity: Ability to accommodate decreases in liability and to fund increases in assets.

The maturity ladder: Comparison of future cash inflows to its future cash outflows over a series of specified time periods.

Alternative scenarios: Evaluating whether a bank has sufficient liquid funds based on the behaviour of cash flows under the different "what if" scenarios.

Crisis Preparedness: Contingency Planning

Strategic Questions

- Does management have a strategy for handling a crisis? Are procedures in place for accessing cash in an emergency?

Components of a Crisis Strategy

– Managerial Coordination: Timely, uninterrupted information flows to senior management.

– **Clear Division of Responsibility:** All personnel must understand their exact roles to avoid confusion and wasted resources.

– Altering Behaviors: Strategies to market **assets more aggressively or raise deposit rates** to retain fleeing funds.

Backup Liquidity: Explicit procedures for making up shortfalls, utilizing unused credit facilities, and drawing on domestic central bank credit lines.

Reserve Bank of India Guidelines for Maturity Buckets: Reserve Bank of India has given a for bucket-wise classification of assets and liabilities to be followed by Indian banks while p Structural Liquidity Statement. These are the guiding factors for the banks. All the assets ar are classified into the following time buckets as given below:

- Tomorrow
- 2–7 days
- 8–14 days
- 15–30 days
- 31 days and up to 2 months
- Over 2 months and up to 3 months
- Over 3 months and up to 6 months
- Over 6 months and up to 1 year
- Over 1 year and up to 3 years
- Over 3 years and up to 5 years
- Over 5 years

Regulatory Framework: RBI Maturity Buckets

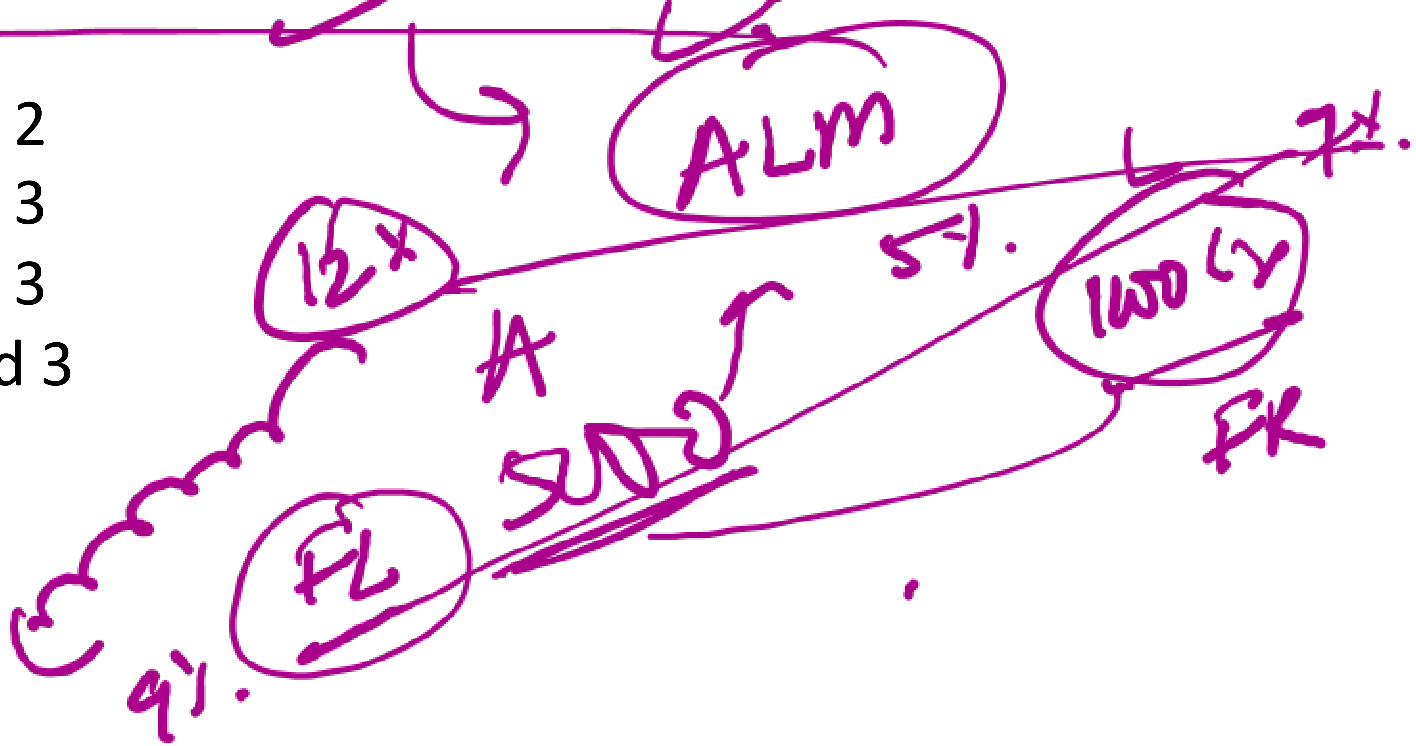
The Reserve Bank of India mandates the following bucket-wise classification for preparing the Structural Liquidity Statement:

1. Tomorrow
2. 2–7 days
3. 8–14 days
4. 15–30 days
5. 31 days and up to 2 months
6. Over 2 months and up to 3 months
7. Over 3 months and up to 6 months
8. Over 6 months and up to 1 year
9. Over 1 year and up to 3 years
10. Over 3 years and up to 5 years
11. Over 5 years

Q. Which of the following statements are correct regarding liquidity risk and management in banks?

1. Liquidity risk is an extraordinary event that arises only during financial crises and rarely impacts day-to-day banking operations.
2. Basel III introduced the Liquidity Coverage Ratio and Net Stable Funding Ratio specifically to address liquidity risk in banks.
3. A bank is considered to have adequate liquidity if it can meet increases in assets and decreases in liabilities promptly and at reasonable cost.

A. Only 1 and 2
B. Only 2 and 3
C. Only 1 and 3
D. All 1, 2, and 3



Answer: B

Solution:

Statement 1 is incorrect: Liquidity risk is a normal aspect of daily bank operations, not just an extraordinary event.

Statement 2 is correct: Basel III introduced LCR and NSFR to manage liquidity risks.

Statement 3 is correct: A bank is said to have adequate liquidity when it can fund increases in assets and accommodate decreases in liabilities effectively and affordably.

Q. ABC Bank, a mid-sized Indian commercial bank with operations in domestic and foreign currency markets, faces a sudden deterioration in liquidity. The situation developed as follows:

- A major borrower defaulted on a ₹300 crore loan, turning the standard asset into an NPA. TR
- Simultaneously, a fraudulent transaction of ₹200 crore was discovered, shaking public trust.
- Due to widespread news of the fraud, several high-value depositors initiated premature withdrawals, and wholesale deposits worth ₹500 crore were not renewed.
- A foreign bank, expecting funds from ABC Bank under a currency swap, failed to receive the payment, impacting its own liquidity chain. CR
- Rumours in the international market led foreign creditors to demand early settlement of USD liabilities, but ABC Bank struggled to mobilise forex funds quickly. F

Which of the following types of liquidity risks are reflected in this scenario?

1. Funding Risk
2. Time Risk
3. Call Risk

A) Only statements 1 and 3 are correct
B) Only statements 2 and 3 are correct
C) Only statements 1 and 2 are correct
D) All statements (1, 2, and 3) are correct

Q. Which of the following **statements regarding liquidity risk management** in banks are correct?

1. Liquidity risk management is only relevant during periods of financial crisis, not in normal operations.
2. A robust liquidity management structure should include good MIS, scenario analysis, and contingency planning.
3. The transformation of illiquid assets into liquid assets is a central function of banking and affects liquidity planning.

A) Only statements 1 and 3 are correct
B) Only statements 2 and 3 are correct
C) Only statements 1 and 2 are correct
D) All statements (1, 2, and 3) are correct

Q. Which of the following statements regarding the treatment of foreign currencies in liquidity management by banks is/are correct?

1. Banks operating in multiple currencies face complexity in liquidity management due to both operational decentralisation and potential information asymmetry in foreign markets.
2. A fully centralised liquidity management approach allows each foreign office to manage its own liquidity independently with no oversight from the head office.
3. If exchange markets cease to function, the bank's capacity to convert domestic funds into foreign currency becomes a critical part of its contingency strategy.

A) Only statements 1 and 3 are correct
B) Only statements 2 and 3 are correct
C) Only statements 1 and 2 are correct
D) All statements (1, 2, and 3) are correct

Answer: A

Solution:

Statement 1 – Correct: Banks dealing with multiple currencies face added complexity due to two main reasons: (i) foreign currency liability holders may react slowly or inaccurately to concerns, and (ii) it may not be feasible to convert domestic liquidity into foreign currency during disruptions.

Statement 2 – Incorrect: In a fully centralised system, the head office manages liquidity across all currencies. Independent control by foreign offices with no oversight refers to a decentralised model, not centralised.

Statement 3 – Correct: If access to exchange markets is disrupted, a bank must rely on its ability to convert domestic funds into foreign currency. This forms a key part of its contingency planning.

Q.Which of the following statements are correct with respect to the setting of tolerance levels and limits for liquidity risk by banks?

1. Banks may be required to set or follow limits such as cumulative cash flow mismatches as a percentage of total liabilities over various time horizons.
2. Liquid assets used for calculating liquidity coverage should be based on book value without discounting, as they are expected to retain value even during stress.
3. Limits on loan-to-capital ratio and loan-to-deposit ratio are among the parameters used to manage liquidity risks.

A) Only statements 1 and 3 are correct
B) Only statements 2 and 3 are correct
C) Only statements 1 and 2 are correct
D) All statements (1, 2, and 3) are correct

Answer: A

Solution:

Statement 1: Correct. Limits on cumulative mismatches over time (next day, week, month, etc.) are part of liquidity risk management.

Statement 2: Incorrect. Liquid assets are evaluated conservatively with discounts to account for price volatility, especially during stress.

Statement 3: Correct. Limits on loan-to-deposit and loan-to-capital ratios help monitor overextension of credit against liabilities and capital.

Q. Identify the correct statements related to the Flow Approach and maturity ladder for liquidity management:

1. Flow approach calculates net funding requirements based on residual maturity of assets and liabilities.
2. A maturity ladder compares future cash inflows and outflows over specified time periods.
3. Negative liquidity gap in a time bucket implies surplus liquidity which can be deployed for lending.

A) Only statements 1 and 3 are correct

B) Only statements 2 and 3 are correct

C) Only statements 1 and 2 are correct

D) All statements (1, 2, and 3) are correct

Answer: C

Solution:

Statement 1: Flow approach involves calculating the difference in cash inflows and outflows based on when they mature.

Statement 2: A maturity ladder helps in placing future inflows and outflows across time buckets to manage liquidity.

Statement 3: Incorrect. A negative gap shows a shortage of funds, not a surplus.

L A

100 90

-10

Q. Which of the following statements regarding Liquidity Risk Management in banks are correct?

1. Effective liquidity management helps a bank avoid selling assets at depressed prices, preserving value under stress scenarios.
2. A bank with strong liquidity may still face a high default risk premium if its capital adequacy is weak.
3. Monitoring macroeconomic and money market trends plays a critical role in managing a bank's liquidity position.
4. A sudden increase in non-performing assets can strain a bank's liquidity position.
5. Selling illiquid assets is the preferred first response to short-term liquidity needs, as they typically yield higher market prices.

LCR

A. Only 1, 2, and 3
B. Only 1, 3, and 4
C. Only 2, 4, and 5
D. Only 1, 3, 4, and 5

Answer: B

Explanation:

Statement 1: Correct. Effective liquidity management allows the bank to avoid asset sales at fire-sale prices, preserving going concern value.

Statement 2: Incorrect. A strong liquidity position and strong capital adequacy both contribute to lower default risk premium; weak capital can raise concerns, but strong liquidity alone doesn't lead to a high premium.

Statement 3: Correct. Monitoring economic and market trends is essential for anticipating and planning liquidity needs.

Statement 4: Correct. An increase in NPAs reduces earnings and repayment inflows, directly affecting liquidity.

Statement 5: Incorrect. Illiquid assets are not preferred for immediate liquidity generation due to difficulty in sale and possible loss in value.

Q. ABC Bank, a mid-sized Indian commercial bank with operations in domestic and foreign currency markets, faces a sudden deterioration in liquidity. The situation developed as follows:

- A major borrower defaulted on a ₹300 crore loan, turning the standard asset into an NPA.
- Simultaneously, a fraudulent transaction of ₹200 crore was discovered, shaking public trust.
- Due to widespread news of the fraud, several high-value depositors initiated premature withdrawals, and wholesale deposits worth ₹500 crore were not renewed.
- A foreign bank, expecting funds from ABC Bank under a currency swap, failed to receive the payment, impacting its own liquidity chain.
- Rumours in the international market led foreign creditors to demand early settlement of USD liabilities, but ABC Bank struggled to mobilise forex funds quickly.

Which of the following types of liquidity risks are reflected in this scenario?

1. Funding Risk
2. Time Risk
3. Call Risk
4. Systemic Risk
5. Instrument-Specific External Risk

A. 1, 2 and 3 only
B. 1, 2, 4 and 5 only
C. 1, 2, 3 and 4 only
D. All of the above (1 to 5)

Answer: D

Explanation:

Funding Risk is present due to premature withdrawals, non-renewal of deposits, and fraud-induced outflows.

Time Risk arises from NPA formation and defaults in expected inflows.

Call Risk occurs when currency swap obligations fail, and non-fund-based exposures crystallize.

Systemic Risk is evident as the bank's failure impacted other banks, and public confidence was lost.

Instrument-Specific External Risk appears due to the complexity in managing forex liabilities, and foreign creditors reacting to market rumours.

Q. Which of the following statements regarding liquidity risk management in banks are correct?

1. Liquidity risk management is only relevant during periods of financial crisis, not in normal operations.
2. A robust liquidity management structure should include good MIS, scenario analysis, and contingency planning.
3. The transformation of illiquid assets into liquid assets is a central function of banking and affects liquidity planning.
4. The Board of Directors need not involve itself in routine monitoring of liquidity risk once policies are in place.
5. A bank should establish a dedicated internal structure, such as an ALCO or risk management department, to manage liquidity as per approved strategy and policies.

A. Only 2 and 3 are correct
B. Only 2, 3, and 5 are correct
C. Only 1, 2, and 4 are correct
D. Only 1 and 4 are correct

Answer: B

Solution:

Statement 1: Incorrect: Liquidity risk management is vital at all times, not just in crisis, because any bank's survival and ability to respond to stress depends on ongoing liquidity.

Statement 2: Correct: Good MIS, scenario analysis, diversification of funding, and contingency planning are explicitly mentioned as crucial for managing liquidity.

Statement 3: Correct: Banks routinely convert illiquid assets into more liquid forms, and this transformation is central to liquidity planning and strategy.

Statement 4: Incorrect: The Board must actively monitor the bank's liquidity risk profile on an ongoing basis; it is not a one-time setup.

Statement 5: Correct: The bank should indeed have a clear internal structure (e.g., ALCO or similar group) responsible for executing liquidity management strategies and policies.

Q. Which of the following statements are correct with **respect to the setting of tolerance levels and limits for liquidity risk by banks?**

1. Banks may be required to set or follow limits such as cumulative cash flow mismatches as a percentage of total liabilities over various time horizons.
2. Liquid assets used for calculating liquidity coverage should be based on book value without discounting, as they are expected to retain value even during stress.
3. Limits on loan-to-capital ratio and loan-to-deposit ratio are among the parameters used to manage liquidity risks.
4. A bank may be required to restrict high reliance on a particular liability category, such as high-cost deposits, by setting flexible percentage caps.
5. An acceptable liquidity mismatch in the 1–14 day and 15–28 day time buckets is around 80% of cash outflows, indicating tighter control in short-term liquidity management.

A. Only 1, 3, and 4 are correct
B. Only 2, 3, and 5 are correct
C. Only 1, 3, 4, and 5 are correct
D. All statements are correct

Answer: C

Solution:

Statement 1: Correct. Limits on cumulative mismatches over time (next day, week, month, etc.) are part of liquidity risk management.

Statement 2: Incorrect. Liquid assets are evaluated conservatively with discounts to account for price volatility, especially during stress.

Statement 3: Correct. Limits on loan-to-deposit and loan-to-capital ratios help monitor overextension of credit against liabilities and capital.

Statement 4: Correct. Flexible limits may be imposed on liability categories like high-cost deposits to avoid over-dependence.

Statement 5: Correct. Banks manage liquidity by maintaining mismatch in certain time buckets (1–14 days, 15–28 days) around 80% of cash outflows.

Q. Identify the correct statements related to **the Flow Approach and maturity ladder for liquidity management:**

1. Flow approach calculates net funding requirements based on residual maturity of assets and liabilities.
2. A maturity ladder compares future cash inflows and outflows over specified time periods.
3. Negative liquidity gap in a time bucket implies ~~surplus~~ liquidity which can be deployed for lending.
4. The maturity ladder assists in identifying excess or deficit of funds at different time intervals.
5. Larger funding gaps in distant periods are easier to manage than those in near-term periods.

A. 1, 2, 4 and 5 only

B. 1, 2 and 4 only

C. 1, 3, 4 and 5 only

D. 2, 3, 4 and 5 only

Answer: A

Solution:

Statement 1: Flow approach involves calculating the difference in cash inflows and outflows based on when they mature.

Statement 2: A maturity ladder helps in placing future inflows and outflows across time buckets to manage liquidity.

Statement 3: Incorrect. A negative gap shows a shortage of funds, not a surplus.

Statement 4: It allows banks to spot when and where they will have excess or shortfall in liquidity.

Statement 5: Gaps in distant periods can be planned for in advance; near-term gaps are harder to manage.

Q. Which of the following statements regarding the evaluation **of cash flows arising from a bank's liabilities and related liquidity management practices is/are correct?**

1. Banks may assume certain core deposits will stay even in a worst-case scenario due to factors like reliance on safety nets and high switching costs.
2. All off-balance sheet items are excluded from liquidity analysis as they are non-cash items and pose no direct liquidity risk.
3. In liquidity stress scenarios, wholesale deposits are typically assumed to run off at the earliest possible maturity due to their volatility.
4. A statistical analysis of demand deposit behaviour, incorporating factors such as seasonality and interest rate sensitivity, is often employed to estimate effective maturities.
5. Credit lines that banks can draw upon during a liquidity crisis are uniformly available and unaffected by clauses such as material adverse change provisions.

A. Only 1, 3 and 4
B. Only 2 and 5
C. Only 1, 3, 4 and 5
D. Only 1 and 4

Answer: A

Solution:

Statement 1 – Correct: Core deposits may be assumed to stay even in adverse conditions due to reliance on deposit insurance or switching costs.

Statement 2 – Incorrect: Off-balance sheet items are considered in advanced liquidity analysis, especially under stress scenarios, as they can cause significant cash drains.

Statement 3 – Correct: Wholesale deposits are volatile and expected to run off early in crisis scenarios.

Statement 4 – Correct: Banks use statistical tools factoring in seasonality, macroeconomics, and interest rates to estimate demand deposit behaviours.

Statement 5 – Incorrect: Credit lines, especially those with material adverse change clauses, may not be reliably available in a crisis.

5 min

UNIT 30 Interest Rate Risk Management

STRUCTURE

Essentials of Interest Rate Risk

Interest Rate Risk (IRR) is the risk to a bank's earnings, capital, and market value arising from **adverse movements in interest rates**. While taking interest rate risk is a **normal and potentially profitable** part of banking, **excessive risk** can threaten a bank's **earnings and solvency**.

Impact of Interest Rate Changes

Earnings Impact

1. Changes in interest rates affect **Net Interest Income (NII)** and **Net Interest Margin (NIM)**.
2. NII volatility arises due to differences in **principal amounts, maturities, and repricing dates** of assets and liabilities (rollover risk).

Market Value Impact

1. Market value of assets and liabilities equals the **present value of future cash flows**.
2. **Rising interest rates → higher discount rates → lower market values**.
3. **Falling interest rates → higher market values**.

Maturity Mismatch Risk

1. If **asset maturities are longer than liabilities**, a rise in interest rates causes the **value of assets to fall more than liabilities**.
2. This can lead to **economic losses** and even **insolvency risk**.

Nature and Measurement

Interest rate risk reflects **volatility in NII or NIM** due to interest rate movements.

Banks increasingly assess the **embedded interest rate risk** in their balance sheets.

Risk Management

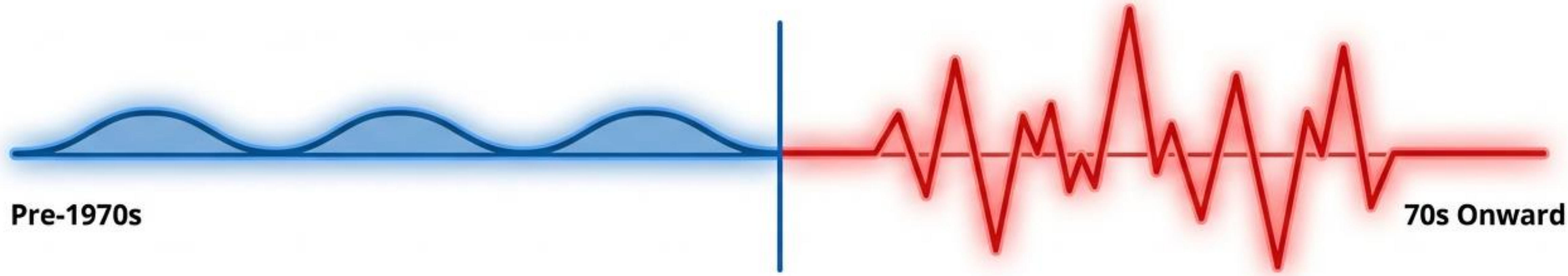
Maintaining interest rate risk within **prudent and acceptable limits** is crucial for **bank safety and soundness**.

Effective risk management processes are essential to control this risk.

Types of Interest Rate Risk

- ✓ **Mismatch (Gap) Risk**
- ✓ **Basis Risk**
- ✓ **Net Interest Position Risk**
- ✓ **Embedded Option Risk**
- ✓ **Yield Curve Risk**
- ✓ **Price Risk**
- ✓ **Reinvestment Risk**

The Evolution of Interest Rate Risk in Banking



The Era of Regulation (Pre-1970s):

Banks operated under mandatory deposit and lending rates, insulated from market pressures. The classic "3-6-3" rule prevailed: "bring in short-term deposits at 3%, lend long at 6%, and be home by 3 p.m." Only credit and liquidity risks were major concerns.

The Catalyst for Change:

Deregulation in the 70s, 80s, and 90s exposed poorly managed interest rate risk globally, leading to bank failures when long-term fixed assets were funded with short-term volatile liabilities.

The Indian Context:

Deregulation of the financial system in India provided operational freedom, leaving the pricing of assets and liabilities to commercial judgement. Earning on assets and cost of liabilities are now closely tied to market volatility.

Defining the Essentials of Interest Rate Risk

Definition: Interest rate risk is the exposure of a bank's financial condition to adverse movements in interest rates. While accepting this risk is a normal part of banking and a source of profitability, excessive risk threatens earnings and capital.

Impact on Net Interest Income (NII): It refers to volatility in Net Interest Income (NII) or variations in Net Interest Margin (NIM) (i.e., NII divided by Earning Assets) due to changes in interest rates. This arises from holding assets and liabilities with different principal amounts, maturity dates, or repricing dates ("rollover rates").

Impact on Economic Value: Changes in interest rates alter the present value of future cash flows, affecting the underlying value of assets, liabilities, and off-balance-sheet (OBS) instruments.

Essentials of Interest Rate Risk

The **market value of assets and liabilities** equals the **present value of their current and future cash flows**.

Rising interest rates increase the discount rate and **reduce market values** of assets and liabilities; falling rates do the opposite. When banks hold **longer-term assets than liabilities**, a rise in interest rates causes the **value of assets to fall more than liabilities**, leading to **economic loss** and possible **insolvency risk**.

Interest Rate Risk is reflected in:

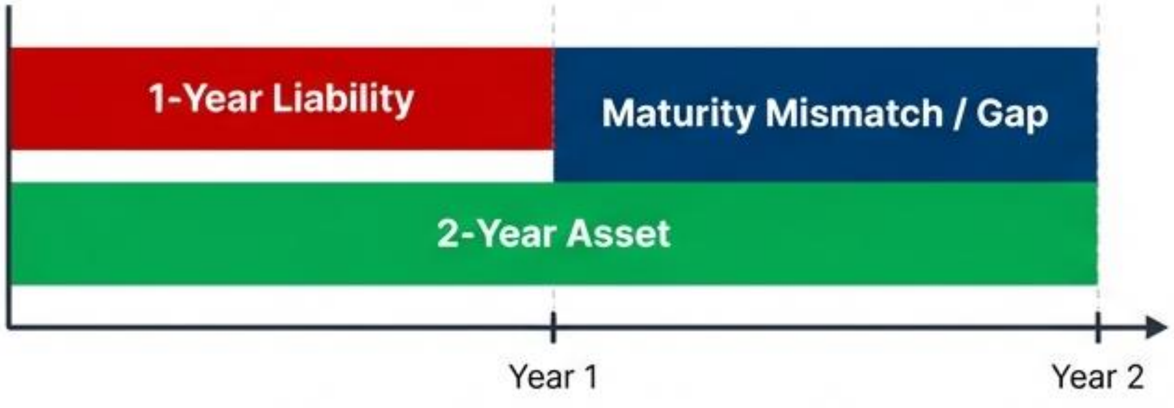
- Volatility of **Net Interest Income (NII)**, or
- Changes in **Net Interest Margin (NIM = NII / Earning Assets)**.

•This risk arises due to mismatches in **principal amounts, maturities, or repricing dates** of assets and liabilities (also called **rollover risk**).

•Effective **interest rate risk management** is essential to maintain bank **safety and soundness**.

•Banks now actively assess interest rate risk as a **drag on profitability** and measure the risk embedded in their balance sheets.

Primary Sources of Risk: Gap and Basis Risk



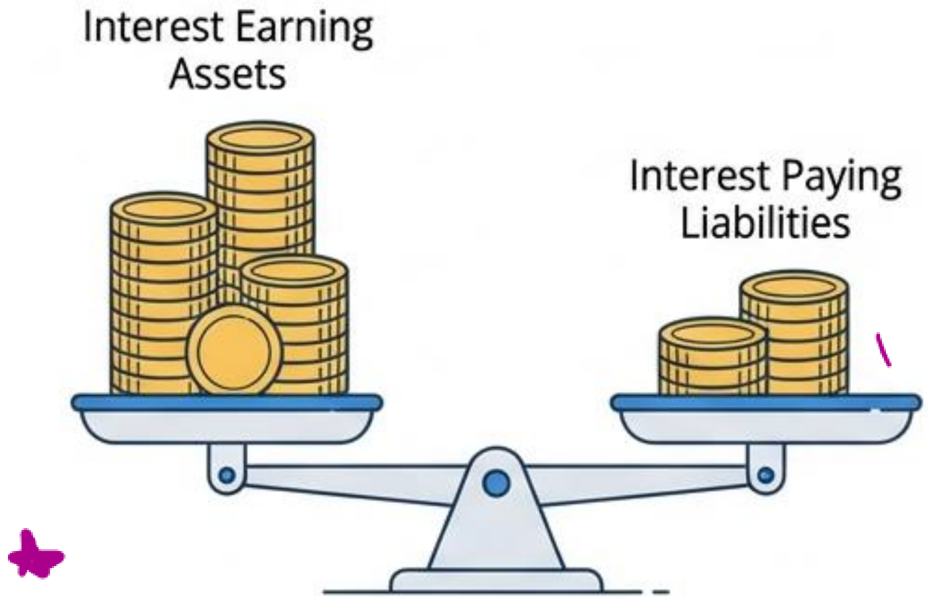
Gap or Mismatch Risk:

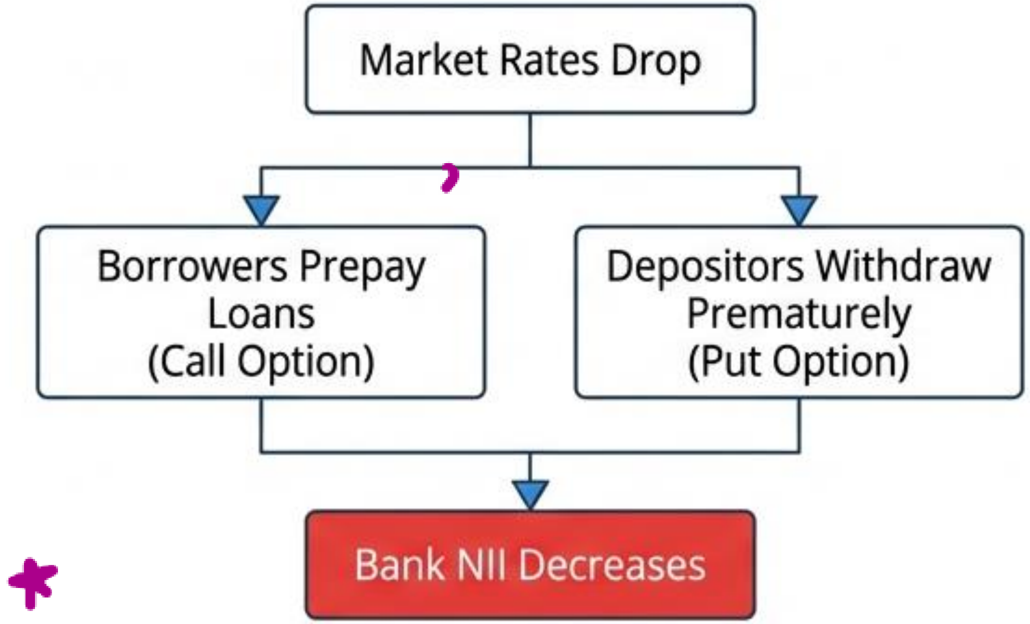
- Arises from holding assets and liabilities with different principal amounts, maturity dates, or repricing dates.
- The "gap" is the difference between the amount of assets and liabilities on which interest rates are reset during a given period.

Basis Risk:

- Even in a perfectly matched gap position, interest rates of two different instruments seldom change by the same degree during the same period.
- The risk that interest rates of different assets and liabilities may change in different magnitudes is called basis risk.

Structural Sources: Net Position and Embedded Options





Net Interest Position Risk:

- Arises when a bank has more assets on which it earns interest than its liabilities on which it pays interest.
- If interest rates earned on assets change while funding costs remain the same, risk materializes. A bank with a positive net interest position experiences reduced NII as rates decline.

Embedded Option Risk:

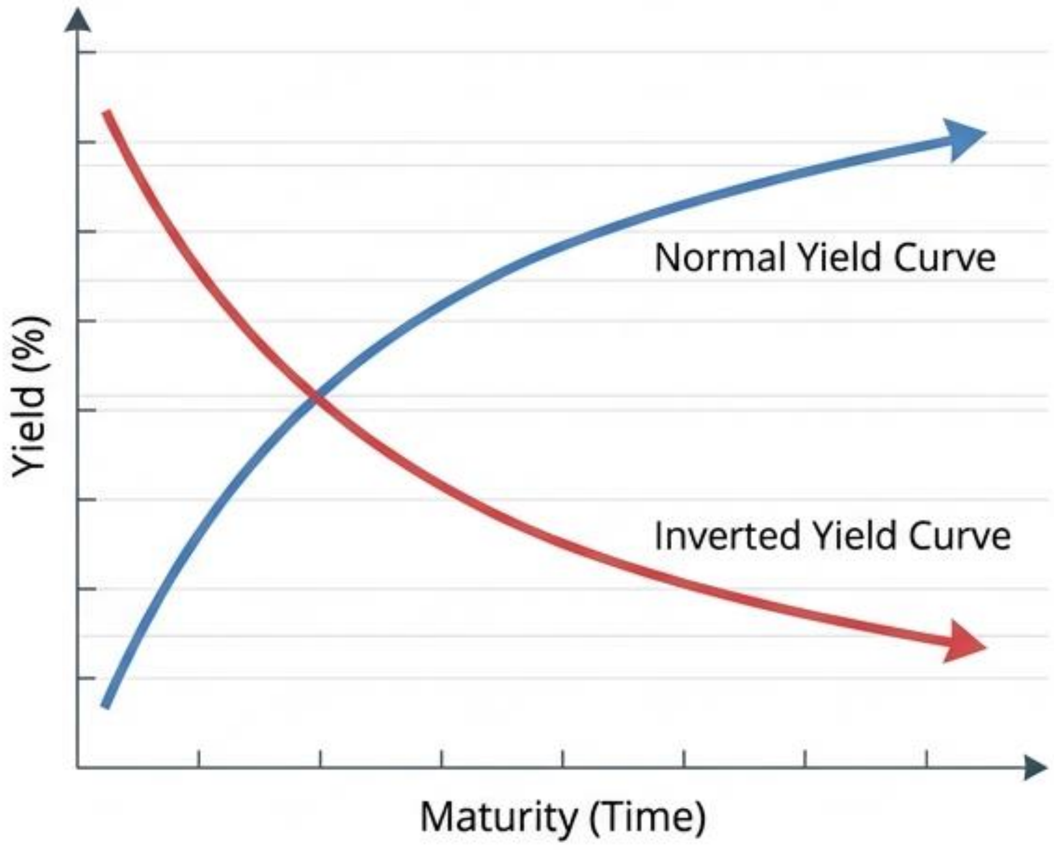
- Large changes in market rates create risk to bank profits through prepayment of loans/bonds (put/call options) and premature withdrawal of deposits before maturity.
- Borrowers naturally pay off loans when interest rates decline, forcing the bank to reinvest at lower yields, thus receiving a lower NII.

Market-Driven Sources: Yield Curve, Price, and Reinvestment Risk

Yield Curve Risk:
Occurs when the yield curve changes its slope and shape from time to time depending upon repricing and various other factors. Changes in the spread relationship between different maturities impact NII.

Price Risk:
Occurs when assets are sold before their maturity dates. Bond prices and yields are inversely related. Primarily associated with the trading book created for short-term profit.

Reinvestment Risk:
Uncertainty regarding the interest rate at which future cash flows can be reinvested. Short-term bonds carry higher reinvestment risk, while long-term bonds carry higher price risk.



Assessing the Damage: Earnings vs. Economic Value Perspectives

B

FX

2

PL

Earnings Perspective

- The traditional approach focusing on the impact of rate changes on accrual or reported earnings.
- Primary focal point is Net Interest Income (NII).
- Reduced earnings threaten financial stability by undermining capital adequacy and market confidence.

Economic Value Perspective

- Focuses on the present value of all expected net cash flows (assets minus liabilities plus OBS positions).
- Provides a comprehensive view of potential long-term effects.
- Economic Value Added (EVA): Shows cost management of businesses working capital; acts as a management incentive ensuring continuity of operations.

Quantifying Exposure: Repricing Schedules and Gap Analysis

Repricing Schedules: Distributes interest-sensitive assets, liabilities, and OBS positions into time bands according to maturity (if fixed-rate) or time remaining to next repricing (if floating-rate).

Gap Analysis: Calculates the 'gap' (RSA - RSL) in each time band. Multiplied by assumed rate changes to yield approximate NII change.

Gap Report

Time Band	RSA (Rate Sensitive Assets)	RSL (Rate Sensitive Liabilities)	Gap (RSA - RSL)
1-28 Days	100	80	+20
29 Days - 3 Months	50	90	-40

Limitations:
- Ignores intra-band variations (assumes all positions mature simultaneously within the band).
- Ignores intra-band variations (assumes all positions mature simultaneously within the band).
- Ignores basis risk (differences in sensitivities of different products).
- Ignores changes in timing of cash flows from embedded options.

Quantifying Exposure: Duration and Modified Duration

Concept:

- Duration is a measure of the percentage change in the economic value of a position that will occur given a small change in the level of interest rates.
- It reflects the timing and size of cash flows that occur before contractual maturity.

Key Rule:

Generally, the longer the maturity or next repricing date, and the smaller the payments occurring before maturity, the higher the duration. Higher duration implies a larger impact on economic value for a given rate change.

Application & Limitations:

- Duration-based weights are used with a maturity/repricing schedule to estimate changes in a bank's economic value.
- Limitations: Estimates from a standard duration approach generally focus on repricing risk and may not reflect basis risk or the non-linearity of price movements during large market shifts.

Advanced Measurement: Static and Dynamic Simulation

Static Simulation:

- Assesses cash flows arising solely from the bank's current on- and off-balance-sheet positions.
- Typically uses straightforward shifts or tilts of the yield curve.

Dynamic Simulation:

- Builds in detailed assumptions about the future course of interest rates and expected changes in the bank's business activity.
- Involves assumptions about bank strategy, customer behavior (e.g., prepayments, deposit withdrawals), and new loans.
- Captures the effect of embedded options better, but runs the risk of becoming a 'black box' that leads to false confidence if underlying assumptions are flawed.

Strategies for Controlling Interest Rate Risk

Core Strategy: The basic strategy focuses on bridging the gap between Rate Sensitive Assets and Rate Sensitive Liabilities, tuning it to interest rate forecasts.

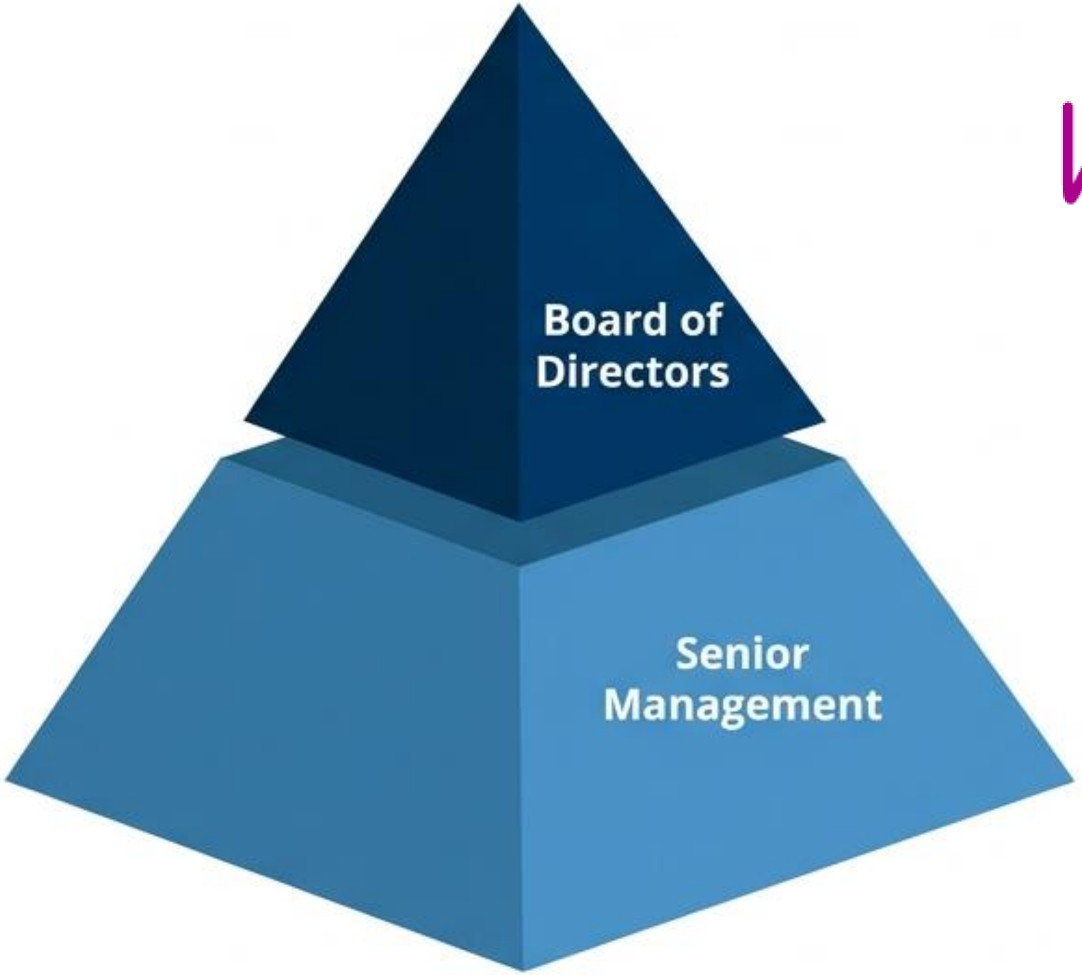
To Reduce Asset Sensitivity
(When rates are expected to fall)

- Extend investment portfolio maturities
- Increase floating rate deposits
- Increase fixed rate lending
- Sell floating rate loans
- Increase short-term borrowings
- Increase long-term lending

To Reduce Liability Sensitivity
(When rates are expected to rise)

- Reduce investment portfolio maturities
- Increase floating rate lending
- Increase long-term deposits
- Increase short-term lending

Governance: Sound Interest Rate Risk Management Practices



Board of Directors:

- Approve strategies and policies with respect to interest rate risk.
- Ensure senior management takes steps to monitor and control risks consistent with approved policies.
- Regularly informed to assess the monitoring and controlling of risk against guidance.

Senior Management:

- Ensure the structure of the bank's business and level of risk are effectively managed.
- Establish appropriate limits on risk taking.
- Maintain adequate systems and standards for measuring risk and valuing positions.
- Implement comprehensive interest rate risk reporting and internal controls.

Regulatory Landscape: RBI Guidelines on IRRBB

What is IRRBB?
Interest Rate Risk in Banking Book refers to current or prospective risk to a bank's capital and earnings arising from adverse rate movements affecting banking book positions.

Capital Requirements:
As part of **Pillar 2** of Basel III Capital regulations, banks must identify risks associated with changing rates on balance sheet and OBS exposures. Excessive IRRBB can threaten a bank's capital base. RBI currently does not permit h[illegible]t holding capital for securities parked in the Banking Book.

Enhanced Governance Guidelines:

- Clearly defined **risk appetite** statement approved by the Board.
- Regular monitoring of nature/level of exposure.
- Systems for prompt management attention and escalations for Board limit violations.
- Identification of IRRBB inherent in all products.

Ensuring Systemic Resilience & Soundness

RBI Guidelines: Measurement and Global Alignment

Measurement Directives: Banks must compute IRRBB and its impact on economic value and earnings based on:

1. Internally selected interest rate shock scenarios addressing the bank's risk profile (ICAAP).

2. Historical and hypothetical interest rate stress scenarios (more severe than normal shocks).

3. The six prescribed interest rate shock scenarios.

Need for Specific Requirements (BCBS Alignment):

- Basel Committee on Banking Supervision (BCBS) finalized standards on IRRBB in April 2016 to promote global consistency and comparability.

- RBI circulated draft guidelines in 2017, requiring banks to implement these standards from April 1, 2019, ensuring Indian banking practices meet global rigorous regulatory benchmarks.

Q. Which of the following statements are correct in the context of interest rate risk and its evolution in the banking system?

1. Before the 1970s, the regulatory framework shielded banks from interest rate risk, making credit and liquidity risk the only major concerns.
2. Interest rate risk became significant after deregulation exposed banks to market-based pricing of assets and liabilities.
3. A bank that funds long-term assets with short-term liabilities is exposed to interest rate risk primarily due to mismatch in maturity and repricing.

A) Only statements 1 and 3 are correct
B) Only statements 2 and 3 are correct
C) Only statements 1 and 2 are correct
D) All statements (1, 2, and 3) are correct

Answer: D

Solution:

Statement 1 – Correct: Till 1970, regulation ensured fixed deposit and lending rates. As a result, banks primarily focused on credit and liquidity risks.

Statement 2 – Correct: Deregulation allowed interest rates to be market-driven, exposing banks to interest rate risk, which had been largely irrelevant before.

Statement 3 – Correct: Holding long-term assets funded by short-term liabilities leads to maturity and repricing mismatches, increasing exposure when rates change.

Q. Which of the following statements are correct in the context of interest rate risk and its evolution in the banking system?

1. Rising interest rates increase the present value of future cash flows from fixed-income assets, thereby increasing the market value of those assets.
2. Effective interest rate risk management focuses on limiting variations in Net Interest Margin (NIM) due to changes in interest rates.

A. Only Statement 1 is correct
B. Only Statement 2 is correct
C. Both statements are correct
D. Neither 1 nor 2 is correct

Answer: B

Solution:

Statement 1 – Incorrect: Rising interest rates decrease, not increase, the present value of fixed future cash flows. Hence, market value of fixed-rate assets declines when interest rates rise.

Statement 2 – Correct: Interest rate risk is reflected in the volatility of Net Interest Income (NII) and Net Interest Margin (NIM). Risk management focuses on stabilizing these.

Q. Bank "Z" has the following interest-sensitive positions on its balance sheet as of June 30:

- Fixed-rate home loans of ₹500 crore maturing in 5 years.
- Floating-rate commercial loans of ₹300 crore repricing every 6 months.
- Fixed-rate deposits of ₹400 crore maturing in 1 year
- Savings deposits of ₹200 crore (assumed to reprice immediately).
- Floating-rate bonds held as assets worth ₹100 crore repricing every 3 months.
- Bank "Z" anticipates a sharp upward movement in interest rates over the next 12 months.

1yr

Which of the following best describes the interest rate risk faced by Bank "Z" and the likely impact due to the expected rate movement?

A. Bank "Z" faces positive gap risk in the short-term bucket, which will benefit the bank if interest rates rise.

B. Bank "Z" faces negative gap risk in the short-term bucket, which will hurt the bank if interest rates rise.

C. Bank "Z" faces positive gap risk in the long-term bucket and is not exposed to immediate interest rate volatility.

D. Bank "Z" faces no gap risk since the aggregate values of assets and liabilities are equal.

Answer: B

Explanation:

Gap analysis (0–1 year bucket):

Interest-sensitive assets (ISA):

Floating-rate commercial loans ₹300 crore (repricing every 6 months)

Floating-rate bonds ₹100 crore (repricing every 3 months)

Total ISA = ₹400 crore

Interest-sensitive liabilities (ISL):

Fixed-rate deposits ₹400 crore (maturing in 1 year)

Savings deposits ₹200 crore (repricing immediately)

Total ISL = ₹600 crore

Gap = ISA – ISL = ₹400 crore – ₹600 crore = -₹200 crore

This is a negative gap, meaning liabilities reprice faster than assets. If interest rates rise, liabilities will cost more, while income from fixed-rate or slowly repricing assets won't increase as fast, hurting the bank.

Q. ABC Bank has a perfectly matched gap position, meaning its assets and liabilities are scheduled to reprice at the same time. However, during a recent shift in the interest rate environment, the interest rate on its fixed-rate term loans increased by 1.75%, while the interest rate on its fixed deposits increased only by 1.25% over the same period.

1.75
1.25

Which of the following best explains the risk ABC Bank is exposed to, and why?

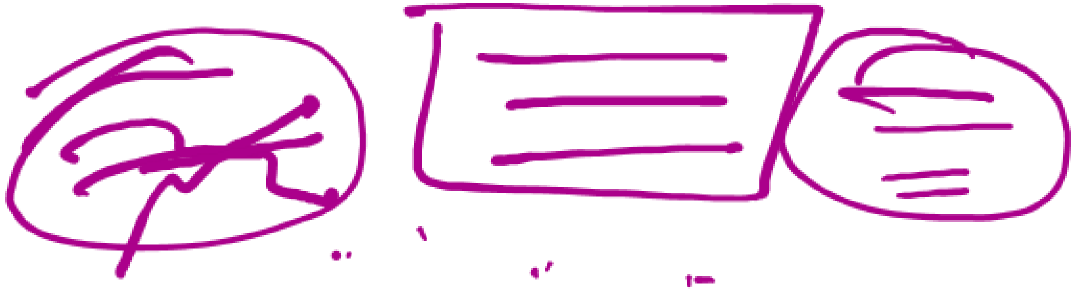
A. Liquidity Risk, because the bank may face a sudden demand for deposits due to misaligned maturity dates.

B. Market Risk, because it is subject to mark-to-market losses on its interest-earning assets.

8.25 8.75

C. Basis Risk, because the change in interest rates on loans and deposits was not proportionate despite matched repricing schedules.

D. Credit Risk, because the borrowers may default due to rising loan rates.



Answer: C

Explanation:

Even though the bank has a matched gap position (same repricing intervals), the rates on assets (loans) and liabilities (deposits) did not move in the same magnitude. The difference in the degree of change (1.75% vs 1.25%) in interest rates between loans and deposits is basis risk, which arises from imperfect correlation in rate movements of different financial instruments.

Q. ABC Bank has a positive net interest position, with a significantly higher volume of interest-earning assets compared to its interest-bearing liabilities. Recently, market interest rates have fallen sharply. During the same period, many of ABC Bank's retail borrowers opted to prepay their fixed-rate home loans, and a noticeable number of depositors prematurely withdrew their fixed deposits, taking advantage of newer, more liquid schemes offered by other banks.

Which of the following risks are ABC Bank likely to be facing under the given scenario, and how would it affect the bank's Net Interest Income (NII)?

1. Net Interest Position Risk — A fall in interest rates will reduce NII due to the structure of ABC Bank's balance sheet.
2. Embedded Option Risk — Prepayment of loans and premature withdrawals reduce interest income and increase reinvestment risk.
3. Credit Risk — Since customers are prepaying loans, the bank is exposed to higher default risk.

A) Only statements 1 and 3 are correct
B) Only statements 2 and 3 are correct
C) Only statements 1 and 2 are correct
D) All statements (1, 2, and 3) are correct

L 5000 Cr
B 3000 Cr → premature
FD
prepay
a y.

Answer: C

Solution:

Statement 1 is correct: A positive net interest position causes NII to fall when interest rates decline.

Statement 2 is correct: Embedded options like prepayment and premature withdrawals reduce income and add reinvestment risk.

Statement 3 is incorrect: Prepayment reduces credit risk (exposure to default), not increases it.

Q. Which of the following statements regarding the **impact of interest rate risk on banks is/are correct?**

1. Under the earnings perspective, interest rate sensitivity is assessed by observing changes primarily in non-interest income, as it directly reflects market fluctuations.
2. The economic value perspective considers the present value of expected net cash flows from assets, liabilities, and OBS positions to evaluate long-term impact of interest rate movements.
3. A rise in market interest rates results in a markdown of the book value of fixed-income investments like government bonds when marked to market, affecting the bank's net worth.
4. Embedded losses refer to the losses recognized immediately in earnings statements when interest rates change unfavorably for unhedged instruments.
5. Economic Value Added (EVA) can be used to assess opportunity cost-adjusted profitability and is sometimes employed as a performance-linked managerial incentive.

A. Only 1, 2, and 3
B. Only 2, 3, and 4
C. Only 2, 3, and 5
D. Only 1, 4, and 5

Answer: C
Solution:
Statement 1 – Incorrect: The earnings perspective primarily focuses on net interest income, not non-interest income. While non-interest income is also considered, the traditional and central component remains net interest income due to its clear link to rate changes.

Statement 2 – Correct: The economic value perspective evaluates sensitivity by calculating the present value of expected cash flows from assets, liabilities, and OBS positions, making it comprehensive.

Statement 3 – Correct: Rising interest rates reduce the market value of existing fixed-rate securities (e.g., bonds), and mark-to-market accounting leads to decline in asset value and net worth.

Statement 4 – Incorrect: Embedded losses are not immediately recognized; they are often unrealized and may be gradually reflected in earnings over time as the instrument matures or is sold.

Statement 5 – Correct: EVA is a valuable tool to measure economic profit and is used in managerial performance incentives. It accounts for actual cost of capital.

Q. Consider the following statements regarding **Measurement of Interest Rate Risk in banks:**

1. Banks must measure interest rate risk to understand the extent of risk exposure and to design appropriate risk management strategies.
2. A bank's interest rate risk measurement system should only focus on its on-balance sheet (OBS) positions, as off-balance sheet exposures are not materially affected.
3. Measurement techniques for interest rate risk range from basic methods like static simulations to advanced dynamic models considering future business activities.

Which of the above statements is/are correct?

A. 1 and 3 only
B. 2 and 3 only
C. 1 and 2 only
D. All of the above

Answer: A

Solution:

Statement 1 is correct: The passage emphasizes that risk must be quantified before it can be managed, which requires an understanding of the extent and position of risk.

Statement 2 is incorrect: The measurement system should cover all material interest rate risks, including those from both assets, liabilities, and off-balance sheet (OBS) positions.

Statement 3 is correct: It is clearly mentioned that measurement techniques vary in complexity, from simple static simulations to sophisticated dynamic models.

Q. Which of the following are valid **characteristics or limitations of Gap Analysis in measuring interest rate risk to earnings?**

1. A liability-sensitive gap implies earnings may fall when interest rates rise.
2. It considers the spread variation between different interest rates in its calculation.
3. It assumes simultaneous repricing of all positions within a time band.
4. A positive gap indicates asset-sensitive exposure.
5. It does not reflect changes in the timing of payments due to interest rate movements.

A. 1, 2 and 3 only

B. 2, 3 and 5 only

C. 1, 3 and 4 only

D. 1, 3, 4 and 5 only

Answer: D

Solution:

Statement 1 is correct: Liability-sensitive gap means higher rates reduce income.

Statement 2 is incorrect: Gap analysis ignores basis risk (spread variation).

Statement 3 is correct: It simplistically assumes all repricing happens together.

Statement 4 is correct: A positive gap suggests income loss when rates fall.

Statement 5 is correct: It does not adjust for timing changes from rate movement.

Q. Identify which of the following statements **are correct with respect to strategies for controlling interest rate risk in banks.**

Liab.

1. Reducing the maturity of the investment portfolio is a strategy to reduce asset sensitivity.
2. Aggressive bankers increase the interest rate gap if they anticipate a rise in interest rates.
3. Liability sensitivity can be reduced by increasing long-term deposits and short-term lending.
4. Risk-averse banks always aim to fully match the repricing of assets and liabilities across all time bands.
5. Using swaps and financial futures allows banks to hedge against exposure to interest rate risk.

A. Only 1, 3 and 4
B. Only 2, 3 and 5
C. Only 1, 2 and 4
D. Only 2, 3 and 4

Answer: B

Solution:

Statement 1 – Incorrect: Reducing investment portfolio maturities is a strategy to reduce liability sensitivity, not asset sensitivity.

Statement 2 – Correct: Aggressive bankers widen the interest rate gap when they expect rates to rise, by repricing assets more frequently than liabilities.

Statement 3 – Correct: Increasing long-term deposits and short-term lending helps in reducing liability sensitivity.

Statement 4 – Incorrect: Risk-averse banks prefer a matched book, but a full match of repricing is neither feasible nor prudent.

Statement 5 – Correct: Banks use derivatives like swaps, futures, options, and FRAs to hedge against interest rate risk.

Q. Which of the following statements correctly describe **sound interest rate risk management practices in banks?**

1. The board of directors should approve interest rate risk management strategies and receive regular updates on the bank's exposure to such risk.
2. Senior management is primarily responsible for executing board decisions but is not expected to maintain any independent risk controls or review processes.
3. Interest rate risk measurement and control functions must be functionally independent from those involved in position-taking activities.
4. A sound interest rate risk management framework requires periodic review of policies and procedures to ensure continued relevance and effectiveness.
5. Interest rate risk reports submitted to senior management should include only aggregated data to maintain confidentiality and avoid excessive operational burden.

A. Only 1, 3, and 4 are correct
B. Only 2, 4, and 5 are correct
C. Only 1, 2, and 5 are correct
D. Only 1, 3, and 5 are correct

Answer: A

Solution:

Statement 1 is correct: The board is expected to approve strategies and receive regular updates on interest rate risk exposure.

Statement 2 is incorrect: Senior management is responsible not just for execution but also for maintaining independent review processes and internal controls.

Statement 3 is correct: Risk measurement and control must be independent of trading functions to avoid conflict of interest.

Statement 4 is correct: Policies and procedures must be periodically reviewed to ensure they remain appropriate and effective.

Statement 5 is incorrect: Reports should contain both aggregate data and granular details to enable proper assessment of sensitivity to risk.

UNIT 31 RAROC and Profit Planning

STRUCTURE

Profit Planning

Banks are **commercial entities** aiming to earn profits while also acting as **financial intermediaries** and **trustees of depositors' funds**.

There is an inherent **conflict** between:

- ✓ Maximizing shareholder profits
- ✓ Protecting depositors' interests and ensuring fixed returns

Hence, banks must operate **prudently and manage risks carefully**.

Profit planning primarily involves **balance sheet management**, covering:

- ✓ Credit activities
- ✓ Investments
- ✓ Non-fund based income

Sources of bank income:

- ✓ **Interest income** (from loans and investments)
- ✓ **Fee-based income**
- ✓ **Treasury income**

The 6 Variables of Profitability

Achieving optimal profitability requires maximizing the first three variables while minimizing the latter three.

Income Variables

1. Interest Income
Derived from lending and securities investments.

2. Fee-Based Income
Traditional (drafts, guarantees, LCs) and modern (cross-selling mutual funds, insurance).

3. Trading Income
Speculative income from FX, equities, and derivatives. Requires stringent controls (e.g., the 1990s collapse of Barings Bank due to Nikkei Futures trading without adequate capitalization).

−

Expense Variables

4. Interest Expenses
Driven by deposit costs. Current and Savings Accounts (CASA) are crucial for securing low-cost funds.

5. Staff Expenses
Managed through productivity improvements and incentive-based packages.

6. Other Operating Expenses
Depreciation, rent, legal fees, and communication charges.

=

Profitability

The Dilemma: Risk vs. Capital (Table 31.1)

Banks cannot simply lend to the highest yielding customer. Basel regulations enforce a risk-weight penalty. As lending risk increases, the required capital also increases, forcing banks to optimize the risk-return tradeoff.

Data Table (Based on an Investment of Rs. 1,000)			
Scenario I: Govt. Securities	6% Yield	0% Risk Weight	Capital Required (@11.50%): 0
Scenario II: AAA Rated Retail	8% Yield	20% Risk Weight	Capital Required (@11.50%): 13.80
Scenario III: AA Rated Corporate	10% Yield	50% Risk Weight	Capital Required (@11.50%): 29.90
Scenario IV: A Rated Corporate	12% Yield	100% Risk Weight	Capital Required (@11.50%): 41.40

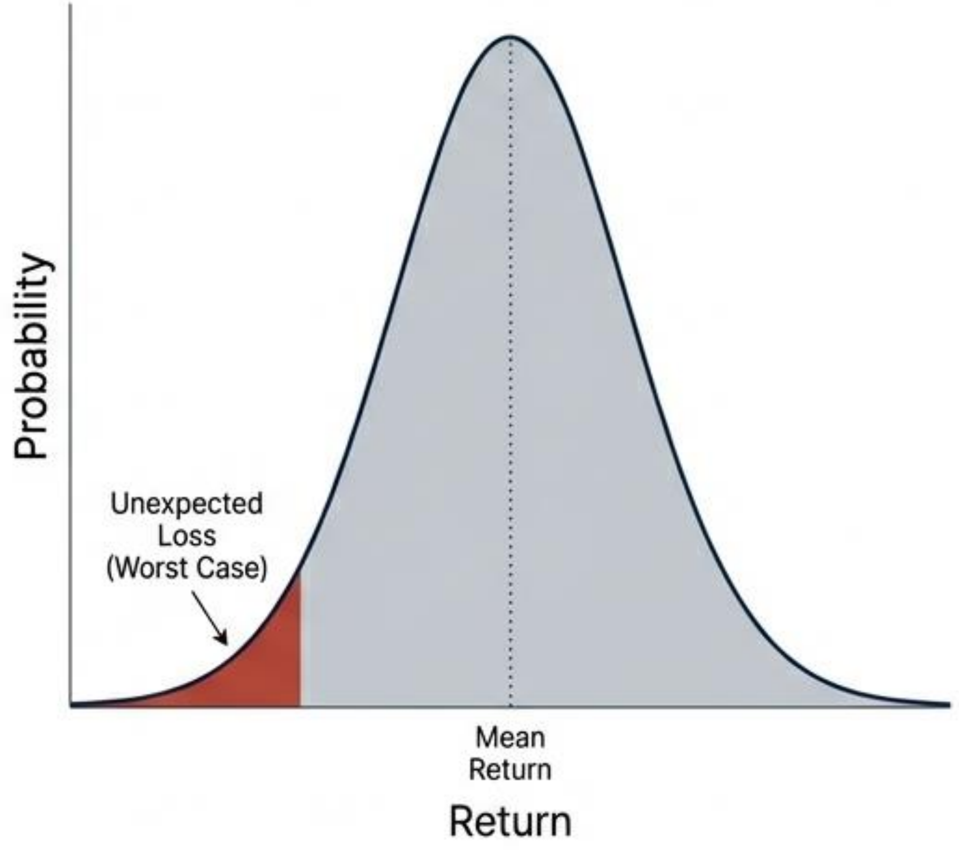
Insight: NPAs compound this issue by creating a two-fold effect: a reduction in generated income and a sudden need for additional capital maintenance.

Concept Deep-Dive: Risk Aggregation

Under Basel guidelines, a bank's Head Office must critically assess internal capital adequacy across all individual lines of business. This requires the summation, or aggregation, of different types of risks.

Measurement Tools:

- **Value at Risk (VaR):** Measures both expected and unexpected losses using worst-case analytical models. Risk is measured in terms of variability of income.
- **Statistical Distribution:** The framework estimates the frequency distribution of return. The Standard Deviation (SD) of this distribution acts as the primary volatility measure.
- **Earnings at Risk (EaR):** An alternative approach often employed to analyze interest rate risk. Rather than moving from the volatility of value through capital, EaR restricts the maximum potential loss to a certain percentage of past/current income based on cash flows.



Higher yield comes with higher risk: Moving from safe government securities to lower-rated corporate lending (AA → A) increases overall portfolio yield (from 6% → 8.6%) but sharply raises **risk-weighted assets** and therefore **capital required**.
Banks must balance **return on capital** — higher-yielding assets demand more capital (especially at higher CAR like 11.5% in Indian context), reducing profitability if capital is scarce.

NPA impact (under Basel II/III): Non-performing assets

- ✓ Reduce interest income
- ✓ Still attract high risk weights → require more capital
- ✓ Double negative effect on profitability

Shift to fee-based income: Traditional interest income is under pressure → banks increasingly rely on **non-interest (fee-based) income** from:

- ✓ Guarantees, letters of credit, remittances, bills, custody, etc.
- ✓ Newer tech-driven services: depository, internet banking, e-commerce, cross-selling (insurance, mutual funds, etc.)
- ✓ Banks evolving into **financial supermarkets**

Treasury Income

Treasury income is the most important income component, earned through trading in:

- ✓ Securities
- ✓ Foreign exchange
- ✓ Equities, bullion, commodities (where permitted)
- ✓ Derivatives

Trading is **speculative in nature** and involves **high risk**, so banks:

- ✓ Maintain strict internal controls
- ✓ Follow regulatory limits (open position limits, MTM valuation, VaR-based explains)

- ➢ While trading can generate **large profits**, it can also cause **heavy losses** if not properly managed.
- ➢ Adequate **capitalisation** is essential to absorb trading losses.
- ➢ The collapse of **Barings Bank** in the 1990s is cited as a warning against unchecked speculative trading.
- ➢ Trading activity should remain **within the bank's risk appetite**.

Expenses of Banks

Bank expenses are mainly of **two types**:

1. Interest Expenses

Arise from interest paid on deposits.

Deposit categories:

- ✓ **Current Deposits** – interest free
- ✓ **Savings Deposits** – low interest; deregulated in India since **25 Oct 2011**, interest paid on daily product basis
- ✓ **Term Deposits** – interest rates largely market-driven

Banks aim to increase **low-cost deposits (CASA)** to reduce interest costs.

Term deposit rates are influenced by:

- ✓ Competition among banks
- ✓ Returns on alternative instruments (debentures, postal deposits, government securities, provident fund, etc.)

2. Operating Expenses

Include:

- ✓ **Staff costs** (salaries, incentives, productivity-linked pay)
- ✓ Other costs: rent, utilities, depreciation, legal expenses, travel, postage, telecom, stationery, etc.

Banks focus on:

- ✓ Improving productivity
- ✓ Linking pay to performance
- ✓ Avoiding wasteful expenditure
- ✓ Rationalising costs through cost-benefit analysis

Determinants of Bank Profitability

Profitability depends on **six key variables**:

1. Interest income
2. Fee-based income
3. Trading income
4. Interest expenses
5. Staff expenses
6. Other operating expenses

- **Maximising the first three** and **minimising the last three** leads to higher profitability.
- All six factors are **inter-dependent**, and optimal balance is essential.

Risk Aggregation and Capital Allocation

Banks develop **internal processes** to assess their **capital needs** based on:

- ✓ Risk profile
- ✓ Business plans
- ✓ Qualitative and quantitative risk factors

Beyond **minimum regulatory capital**, banks must evaluate:

- ✓ **Internal capital adequacy**
- ✓ **Future capital needs** for different business lines and products

The Basel framework emphasises that **capital adequacy relative to economic risk** is vital for long-term bank soundness.

Concept Deep-Dive: Economic Capital

Economic Capital (often called Risk Capital) is the exact measure of capital required to support a bank's financial risk. It acts as the ultimate buffer against future, unexpected losses.

Why Traditional Metrics Fail:

Historically, bank performance was measured by simple yardsticks:

- ROA (Return on Assets): Adjusts profits for the associated book value of assets.
- ROE (Return on Equity): Adjusts profits for the associated book value of equity.

The Flaw: ROA and ROE are unsatisfactory for evaluating modern business lines because they completely ignore the underlying risks taken to achieve those returns. Economic capital forces banks to account for risk in performance evaluation.

ROA + ROE

Concept Deep-Dive: RAROC Methodology

Risk-Adjusted Return on Capital (RAROC) allows all business streams of a financial institution to be evaluated on an equal footing by matching revenues, costs, and risks over a defined time period. The RAROC methodology follows three mandatory steps:

1. Risk Management

Involves the direct measurement of portfolio exposure, the volatility of the assets, and the correlations of the risk factors.

2. Capital Allocation

Requires the choice of a specific confidence level and time horizon for the VaR measure. This mathematical choice translates directly into the required economic capital.

3. Performance Measurement

The final step requires the adjustment of performance to account for the risk capital deployed.

RAPM in Action: The Tale of Two Traders

Risk-Adjusted Performance Measures (RAPM) reveal true performance. Consider two traders who both returned a raw profit of $10 Million. Who performed better?

FX Trader

- Notional Amount: $100 Million
- Volatility: 12% per annum

Calculating Risk Capital (VaR at 99% level over a year):

$$RC = VaR = \$100{,}000{,}000 \times 0.12 \times 2.33$$
$$= \$28 \text{ Million}$$

Bond Trader

- Notional Amount: $200 Million
- Volatility: 4% per annum

Conclusion: RAPM is calculated as Profit divided by Risk Capital (Profit / RC). The Bond Trader is actively performing better than the FX Trader, as their lower volatility activity requires significantly less risk capital to secure the same $10 Million profit.

Performance Measurement: Economic Value Added (EVA)

Economic Value Added focuses on the absolute creation of value during a particular period in excess of the required return on capital. It measures the residual economic profit of a business line or product.

$$\text{EVA} = \text{Profit} - (\text{Capital} \times k)$$

- **Profit:** The generated revenue minus expenses.
- **Capital:** The cost of economic capital allocated to the specific activity.
- **k:** The defined discount rate.

Insight: Assuming the entire worth of a product or project is captured by the EVA framework, a higher EVA directly translates to a better, more viable product for the bank.

Approach 1: RAROC (Risk-Adjusted Return on Capital)

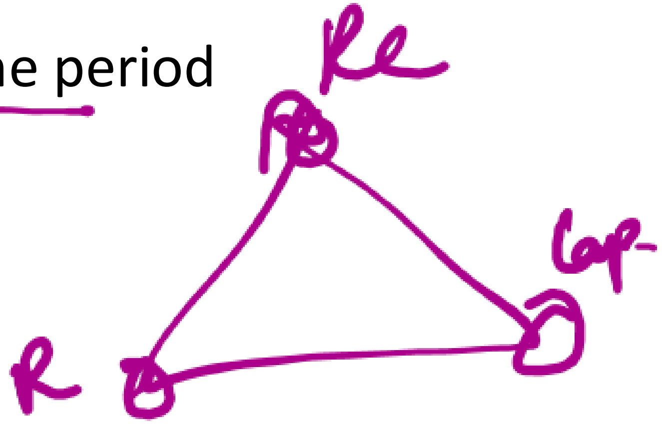
RAROC is the most widely used method for risk aggregation and capital allocation.
Enables **comparison of all business lines on a uniform basis**.

Key features:

- ✓ Differentiates between **expected losses** and **unexpected losses**
  - ✓ Expected losses → covered by provisions/reserves
  - ✓ Unexpected losses → require capital allocation
- ✓ Uses **VaR / worst-case models** to estimate risk
- ✓ Matches **revenues, costs, and risks** over a defined time period

Risk is measured through **variability of income**, using:

- ✓ Frequency distribution of returns
- ✓ Standard deviation as a measure of volatility



- ➢ Capital is allocated in proportion to **risk/volatility**.
- ➢ Riskier activities must generate **higher expected returns**.
- ➢ Thus, risks are **aggregated, priced, and managed** through capital allocation.

Approach 2: EaR (Earnings at Risk)

Similar to RAROC but focuses on:

- ✓ **Cash flows**
- ✓ **Variability in earnings**, rather than capital allocation

Commonly used to analyse **interest rate risk**.

Key aspects:

- ✓ Uses historical data to estimate return distributions
- ✓ Extreme outcomes measured using:
  - ✓ Worst-case scenarios, or
  - ✓ Statistical measures like SD / 2 or SD / 2.69
  - ✓ Banks set limits on **maximum potential loss** as a percentage of income or market value

Unlike RAROC:

- ✓ Directly links risk to **current earnings impact**
- ✓ Ignores valuation changes in assets and liabilities due to interest rate movements
- ✓ Relies on **subjective assumptions** about risk scenarios

Concept Deep-Dive: Capital Allocation

Capital is allocated to activities as a direct function of risk and volatility. The risky position must carry an expected rate of return on the allocated capital that compensates the bank for the associated incremental risk.

Expected Losses

These are the anticipated costs of doing business. The pricing of products should inherently provide a buffer against expected losses. They are covered by specific reserves and provisions on the balance sheet.

Unexpected Losses

These are extreme, unanticipated financial hits. Unexpected losses require direct capital allocation. The exact amount of capital allocated is determined by specific mathematical principles:

- Confidence Levels
- Time Horizon
- Diversification and Correlation

Economic Capital and RAROC:

Core Concepts

Expected Loss → Amount banks must reserve/provide for (through pricing) to cover anticipated future losses.

Unexpected Loss → The buffer of **economic capital** (also called **risk capital**) needed to absorb rare but severe losses. This is the true risk-based capital banks must hold to remain solvent.

Activities/products/business lines vary in how much **risk capital** they consume → those requiring more risk capital demand **higher returns** to be worthwhile.

Risk-Adjusted Return on Capital (RAROC)

RAROC (and related RAPM family of measures) evaluates performance by dividing profit by the **economic/risk capital** used — not just by assets or equity.

Goal: Create risk-adjusted benchmarks to assess true economic value added across:

- ✓ Individual transactions/products
- ✓ Customer relationships
- ✓ Business lines/units
- ✓ Entire bank

Helps decide compensation, resource allocation, and which activities to expand.

Why traditional measures fall short

- ✓ **ROA** (Return on Assets) — adjusts profit for book value of assets → ignores risk.
- ✓ **ROE** (Return on Equity) — adjusts profit for book equity → also ignores risk differences.

Neither accounts for the varying risk levels across activities → can mislead performance evaluation.

Practical Example (FX vs Bond Trader)

Two traders each earn **$10 million profit**:

A

FX Trader: Notional = $100 million, annual volatility = 12% → Risk Capital (VAR at 99% confidence, 1-year, normal distribution) ≈ $100M × 12% × 2.33 ≈ **$28 million**

→ RAPM = $10M / $28M ≈ **35.7%**

B

Bond Trader: Notional = $200 million, annual volatility = 4% → Risk Capital ≈ $200M × 4% × 2.33 ≈ **$18.64 million** → RAPM = $10M / $18.64M ≈ **53.6%**

RAROC Methodology – Key Components

RAROC (Risk-Adjusted Return on Capital) integrates **risk management**, **capital allocation**, and **performance measurement**:

- ✓ **Risk Management** — Measures portfolio exposure, including volatility and correlations across risk factors (credit, market, operational, etc.).
- ✓ **Capital Allocation** — Involves selecting a confidence level (e.g., 99%) and time horizon for Value at Risk (VaR) or similar measures, converting unexpected loss into **economic capital** (risk capital) that the bank must hold as a buffer.
- ✓ **Performance Measurement** — Adjusts reported profits/performance for the **risk capital** consumed, ensuring returns reflect the true risk taken (not just raw profit).

This creates benchmarks for evaluating products, transactions, customers, business lines, or the entire bank on a risk-adjusted basis.

Link to Economic Value Added (EVA)

EVA measures **residual economic profit** after deducting the cost of economic capital: **EVA = Profit – (Capital × k)** where *k* is the cost of capital (discount/hurdle rate, e.g., weighted average cost of capital or required return).

- ✓ Higher EVA indicates better value creation for shareholders.
- ✓ RAROC and EVA are closely related — both focus on **risk-adjusted profitability** and economic value over traditional metrics like ROA/ROE (which ignore risk differences).

The Macro View: The Indian Banking Industry

Historically, Indian banking relied heavily on regulated interest rates and directed investment. Starting with the Narasimham Committee report of 1991, the sector underwent massive deregulation. Today, profitability and capital adequacy are the most critical parameters of a bank's functioning.

The modern competitive landscape features diverse ownership patterns:

- Public Sector Banks
- New Generation Private Sector Banks
- Old Generation Private Sector Banks
- Co-operative Banks & Regional Rural Banks
- Differentiated Banks: Small Finance Banks and Payment Banks (recently permitted by the RBI to open specific market niches).

RAROC and Profit Planning in the Indian Banking Context

Indian banking has transformed since the early 1990s (post-**Narasimham Committee Report, 1991**), moving from:

- ✓ Regulated interest rates and directed credit → deregulated, market-driven environment.
- ✓ Heavy government control → prudential norms, disclosure requirements, and competition.

RBI now emphasizes **prudential regulations**, capital adequacy, and risk management over direct controls.

Current Structure of Indian Banking Sector (as described):

- ✓ Public Sector Banks
- ✓ New Generation Private Sector Banks
- ✓ Old Private Sector Banks
- ✓ Co-operative Banks
- ✓ Regional Rural Banks
- ✓ Small Finance Banks
- ✓ Payment Banks (Differentiated Banks)

Intense competition drives:

- ✓ Technology upgrades
- ✓ Introduction of new products
- ✓ Aggressive marketing
- ✓ Focus on **profitability** as a critical parameter

Special Entity Focus: NaBFID

The financial landscape in India recently saw the birth of a new entity specifically designed for long-term infrastructure financing.

DFI

National Bank for Financing Infrastructure and Development (NaBFID)

- **Establishment:** The NaBFID Act, 2021 received Presidential assent on March 28, 2021, and was enforced on April 19, 2021.
- **Classification:** It is set up as a Development Financial Institution (DFI).
- **Operational Difference:** Unlike standard commercial banks, DFIs do not accept deposits from the general public.
- **Funding Sources:** NaBFID sources its funds directly from the market, the government, and multi-lateral institutions, often supported through government guarantees.

NaBFID – National Bank for Financing Infrastructure and Development

•**Establishment**: Set up as a specialized **Development Financial Institution (DFI)** under the **National Bank for Financing Infrastructure and Development (NaBFID) Act, 2021**.

- Received Presidential assent on **28 March 2021**.
- Enforced/operational from **19 April 2021**.

•**Purpose**: To support and accelerate **long-term infrastructure financing** in India, addressing gaps in non-recourse, long-duration funding for infrastructure projects.

- Marks the introduction of a dedicated new entity in India's financial market focused on infrastructure.

Key Characteristics of NaBFID (as a DFI)

•Unlike commercial banks, DFIs like NaBFID **do not accept public deposits** from individuals.

•Funds are sourced from:

✓ Capital/debt markets (e.g., bonds, borrowings).

✓ Government support.

✓ Multilateral institutions (e.g., World Bank, New Development Bank).

•Often backed by **government guarantees** to enhance credibility and lower funding costs.

Q. Which of the following statements is/are correct regarding the profitability and income-expenditure management of banks?

1. A higher proportion of investment in government securities generally leads to improved interest income due to their lower risk and higher yields compared to credit portfolios.
2. Treasury income, though potentially high-yielding, is speculative in nature and can lead to significant losses if not supported by adequate capital and internal controls.
3. Non-Performing Assets (NPAs) negatively impact both income generation and capital adequacy due to income loss and higher risk weights under Basel norms.

A) Only statements 1 and 3 are correct
B) Only statements 2 and 3 are correct
C) Only statements 1 and 2 are correct
D) All statements (1, 2, and 3) are correct

Answer: B

Solution:

Statement 1 – Incorrect: While government securities are low-risk, they yield lower returns, not higher, compared to credit portfolios—especially lower-rated ones.

Statement 2 – Correct: Treasury income is speculative and risky; internal controls and adequate capital are essential.

Statement 3 – Correct: NPAs reduce interest income and require additional capital under Basel due to higher risk weights.

Q. Which of the following statements are correct in the context of Risk Aggregation and Capital Allocation in banks?

Prov. → Cap.

1. Banks using the RAROC approach differentiate between expected and unexpected losses, allocating capital primarily to cover expected losses.
2. Risk Adjusted Return on Capital (RAROC) enables banks to assess performance across different business lines using a standardized measure.
3. Under the RAROC approach, the capital allocation is linked to the standard deviation of the frequency distribution of returns.

A) Only statements 1 and 3 are correct
B) Only statements 2 and 3 are correct
C) Only statements 1 and 2 are correct
D) All statements (1, 2, and 3) are correct

Answer: B

Explanation:

Statement 1 is incorrect: Under RAROC, expected losses are covered by provisions/reserves, while capital allocation is for unexpected losses.

Statement 2 is correct: RAROC allows comparison across different business streams by standardizing risk-adjusted return.

Statement 3 is correct: RAROC allocates capital based on the volatility (standard deviation) of the return distribution.

Q. Which of the following statements is/are correct regarding the profitability and income-expenditure management of banks?

1. A higher proportion of investment in government securities generally leads to improved interest income due to their lower risk and higher yields compared to credit portfolios.
2. Treasury income, though potentially high-yielding, is speculative in nature and can lead to significant losses if not supported by adequate capital and internal controls.
3. Non-Performing Assets (NPAs) negatively impact both income generation and capital adequacy due to income loss and higher risk weights under Basel norms.
4. CASA deposits are a strategic tool for banks to reduce interest expenses and improve net interest margins.
5. Profitability can be optimised solely by increasing interest income and trading income, without focusing on operating expenses.

A. Only 1, 3, and 4
B. Only 2 and 5
C. Only 1, 2, and 5
D. Only 2, 3, and 4

Answer: D

Solution:

Statement 1 – Incorrect: While government securities are low-risk, they yield lower returns, not higher, compared to credit portfolios—especially lower-rated ones.

Statement 2 – Correct: Treasury income is speculative and risky; internal controls and adequate capital are essential.

Statement 3 – Correct: NPAs reduce interest income and require additional capital under Basel due to higher risk weights.

Statement 4 – Correct: CASA deposits (Current and Savings) are low-cost sources of funds and help reduce overall interest expense.

Statement 5 – Incorrect: Profitability cannot be optimised solely through income increase; operating expenses (staff and other costs) also need to be controlled.

Q. Which of the following statements are correct in the context of Risk Aggregation and Capital Allocation in banks?

1. Banks using the RAROC approach differentiate between expected and unexpected losses, allocating capital primarily to cover expected losses.
2. Risk Adjusted Return on Capital (RAROC) enables banks to assess performance across different business lines using a standardized measure.
3. Under the RAROC approach, the capital allocation is linked to the standard deviation of the frequency distribution of returns.
4. The EaR (Earnings at Risk) model focuses on estimating loss through asset-liability valuation changes caused by market interest rate movements.
5. While RAROC emphasizes capital allocation based on risk volatility, EaR focuses more directly on cash flow variability and earnings implications.

A. Only 2 and 5
B. Only 1, 3, and 4
C. Only 2, 3, and 5
D. Only 2, 3, and 4

Answer: C

Explanation:

Statement 1 is incorrect: Under RAROC, expected losses are covered by provisions/reserves, while capital allocation is for unexpected losses.

Statement 2 is correct: RAROC allows comparison across different business streams by standardizing risk-adjusted return.

Statement 3 is correct: RAROC allocates capital based on the volatility (standard deviation) of the return distribution.

Statement 4 is incorrect: EaR focuses on earnings/cash flows, not valuation changes in assets/liabilities due to interest rates.

Statement 5 is correct: RAROC is capital-focused; EaR is more earnings and cash flow-based.

Q. Which of the following statements are correct **with reference to the concepts of Economic Capital and RAROC?**

1. RAROC provides a more risk-sensitive measure of performance compared to traditional measures like ROA and ROE.
2. Risk capital, used in RAROC, accounts for only market risk and ignores operational and credit risks.
3. The bond trader, despite trading larger volumes, may demonstrate better risk-adjusted returns than the FX trader due to lower volatility.
4. EVA is calculated as: Profit minus (Capital multiplied by discount rate), and reflects value creation over and above the required return on capital.
5. In RAROC methodology, the process of capital allocation is independent of the chosen confidence level and time horizon.

A. Only 1, 2, and 4
B. Only 1, 3, and 4
C. Only 2, 3, and 5
D. Only 1, 3, 4, and 5

Answer: B

Explanation:

Statement 1 – Correct: RAROC adjusts for risk, unlike ROA and ROE which are insensitive to risk.

Statement 2 – Incorrect: Risk capital under RAROC should consider credit risk, operational risk, and interactions, not just market risk.

Statement 3 – Correct: Bond trader requires less risk capital due to lower volatility, yielding a better RAPM.

Statement 4 – Correct: EVA = Profit – (Capital × k) is the correct formula and reflects value creation above capital cost.

Statement 5 – Incorrect: Capital allocation does depend on the confidence level and time horizon, as these define the VAR and thus the economic capital.

Q. Which of the following statements regarding **the evolving banking landscape and financial institutions in India are correct?**

1. Deregulation in the Indian banking sector began after the recommendations of the Narasimham Committee in 1991.
2. Small Finance Banks and Payment Banks are categorized as Differentiated Banks under the RBI framework.
3. Development Financial Institutions (DFIs), like NaBFID, function similar to regular commercial banks by accepting deposits from the public.
4. Mergers in public sector banks include the amalgamation of Bharatiya Mahila Bank and associate banks with the State Bank of India.
5. Indian banks are increasingly focusing on profitability, innovation in products, and technological upgradation due to heightened competition.

A. 1, 2, and 4 only
B. 1, 3, and 5 only
C. 2, 4, and 5 only
D. 1, 2, 4, and 5

Answer: D

Solution:

Statement 1 – Correct: Deregulation and reforms in Indian banking began post-1991 with the Narasimham Committee's recommendations.

Statement 2 – Correct: RBI introduced Small Finance Banks and Payment Banks under the category of "Differentiated Banks."

Statement 3 – Incorrect: DFIs like NaBFID do not accept public deposits; they raise funds from markets, government, and institutions.

Statement 4 – Correct: SBI's associate banks and Bharatiya Mahila Bank have been merged into SBI.

Statement 5 – Correct: Due to competitive pressures, banks are now focusing on profitability, technology upgrades, and innovative products.

Q. A company has the following financial information for the year ended 31 March 2026:

- Capital Employed = ₹850 crore
- EBIT = ₹168 crore
- Debt–Equity Ratio (market value basis) = 3 : 2
- Pre-tax Cost of Debt = 9.2% p.a.
- Beta = 1.25
- Risk-free Rate (10-year G-Sec yield as on 25 May 2026) = 6.9%
- Expected Market Return = 13.7%
- Corporate Tax Rate = 25%

What is the Economic Value Added (EVA) of the company?

A) ₹22.11 crore

B) ₹27.53 crore

C) ₹31.86 crore

D) ₹38.45 crore

Answer: D

Solution:

Step 1: Calculate Cost of Equity using CAPM: Cost of Equity = Risk-free Rate + Beta × (Market Return − Risk-free Rate) = 6.9% + 1.25 × (13.7% − 6.9%) = 6.9% + 1.25 × 6.8% = 6.9% + 8.5% = 15.4%

Step 2: Calculate After-tax Cost of Debt: After-tax Cost of Debt = 9.2% × (1 − 0.25) = 9.2% × 0.75 = 6.9%

Step 3: Calculate WACC: Debt : Equity = 3 : 2

Debt Weight = 3/5 = 0.60

Equity Weight = 2/5 = 0.40

WACC = (0.60 × 6.9%) + (0.40 × 15.4%) = 4.14% + 6.16% = 10.30%

Step 4: Calculate NOPAT: NOPAT = EBIT × (1 − Tax Rate) = 168 × 0.75 = ₹126 crore

Step 5: Calculate Capital Charge

Capital Charge = Capital Employed × WACC = 850 × 10.30% = ₹87.55 crore

Step 6: Calculate EVA: EVA = NOPAT − Capital Charge = 126 − 87.55

= ₹38.45 crore

JAIIB CAIIB

PREP

@JAIIBCAIIBPREPOFFICIAL

JAIIB CAIIB PREP

@JAIIBCAIIBPREPOFFICIAL